KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 9 STANDARD

CHAPTER 11

YEAR 9 MATHEMATICS

Coordinate geometry and graphs

10 lessons · four pages each · Easy drill on number planes, then Medium and worded problems

11.1 The length of an interval

11.2 The midpoint of an interval

11.3 The gradient of an interval

11.4 Graphing linear equations

11.5 The gradient–intercept equation

11.6 Finding the equation of a line

11.7 Solving linear equations graphically

11.8 Linear modelling

11.9 Graphing parabolas

11.10 Parabolas in real life

NAME _____ CLASS _____ TEACHER _____

Artwork: silver seahorse tails deep in Seahorse Valley, coloured in purple and silver, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **four pages**. Read the **learning intention**, copy your teacher's notes into the **Notes** box, and tick each **success criterion** when you can do it.

2. Pages 1, 2 and 3 are **Easy**: small steps, one round at a time. Page 4 is **Medium**, then three **worded problems**.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. The **?** on each diagram shows what to find. Write the **formula**, **substitute** the numbers, then the **answer** with its unit.

3. Some Medium questions come from earlier lessons. The tag (like **11.2** or **Year 8**) says which one.

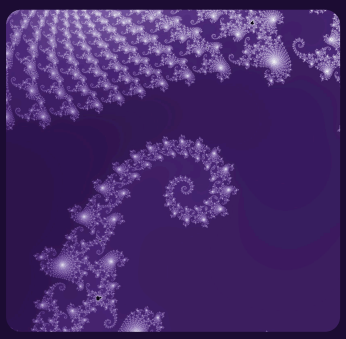
4. Stuck? Read the **Remember** line in the blue box, then look back at your notes and the example. Lengths: **one decimal place** unless the question says otherwise.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
11.1	The length of an interval	3		☐
11.2	The midpoint of an interval	7		☐
11.3	The gradient of an interval	11		☐
11.4	Graphing linear equations	15		☐
11.5	The gradient–intercept equation	19		☐
11.6	Finding the equation of a line	23		☐
11.7	Solving linear equations graphically	27		☐
11.8	Linear modelling	31		☐
11.9	Graphing parabolas	35		☐
11.10	Parabolas in real life	39		☐

In this chapter you will

- find the length, midpoint and gradient of an interval on the number plane
- graph straight lines from a table of values, from intercepts and from $y = mx + c$
- find the equation of a straight line from its graph or from two points
- solve linear equations by reading a graph
- use straight-line graphs and equations to model real situations
- graph parabolas and use them in real situations



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is silver seahorse tails deep in Seahorse Valley, coloured in purple and silver. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

11.1 The length of an interval

EASY

Learning intention We are learning to find the length of an interval on the number plane.

Success criteria
- ☐ I can read the coordinates of the endpoints of an interval
- ☐ I can use the distance formula to find its length
- ☐ I can give a length as a whole number or correct to one decimal place

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

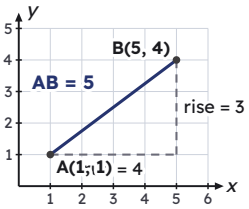
interval: part of a line between two endpoints
run: the change in x: $x_2 - x_1$
rise: the change in y: $y_2 - y_1$
length (d): the distance from one endpoint to the other
(x_1, y_1), (x_2, y_2): the two endpoints

FORMULAS

$d^2 = \text{run}^2 + \text{rise}^2$ (Pythagoras)
$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$
then $d = \sqrt{(\ldots)}$

STEPS

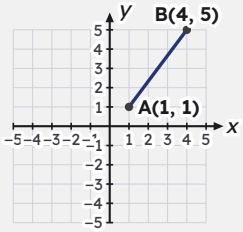
1. write the two points
2. subtract the x-values and the y-values
3. square and add
4. square root, round to 1 d.p.



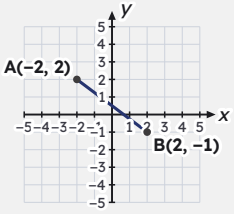
WE DO — With your teacher: copy what they write on the board.

Example 1 Find the length of each interval AB. Round to one decimal place where needed.

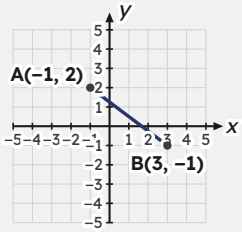
a Length



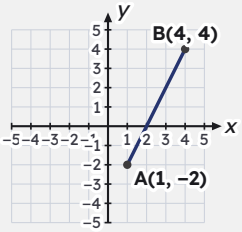
b Length



c Length



d Length



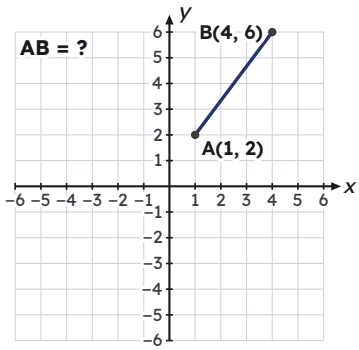
YOU DO — On your own: work down the box, one round at a time. Show your working.

2 QUESTIONS

REMEMBER Subtract the x-values, subtract the y-values (same order) · square both and add · square root last

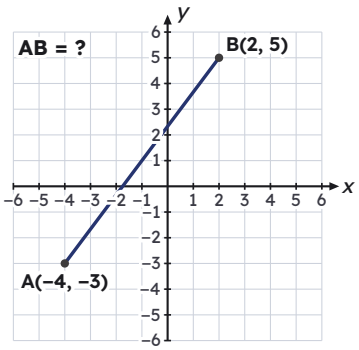
ROUND 1 — Questions 1–2: find the length of AB.

1



Formula $d^2 = (x_2-x_1)^2 + (y_2-y_1)^2$
Substitute _____
Answer _____

2



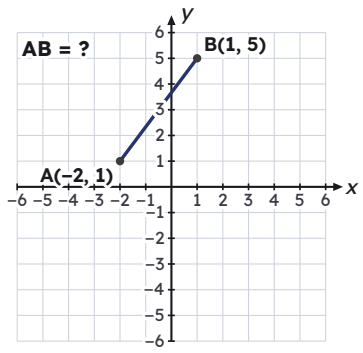
Formula $d^2 = (x_2-x_1)^2 + (y_2-y_1)^2$
Substitute _____
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER A negative squared is positive: $(-3)^2 = 9$ · round to one decimal place at the end.

ROUND 2 Questions 3–4: find the length of AB.

3

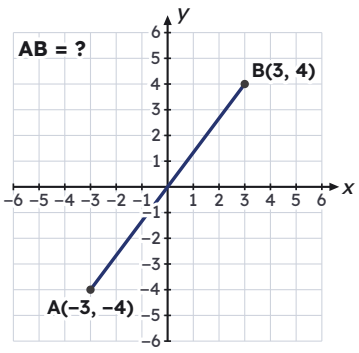


Formula _____

Substitute _____

Answer _____

4



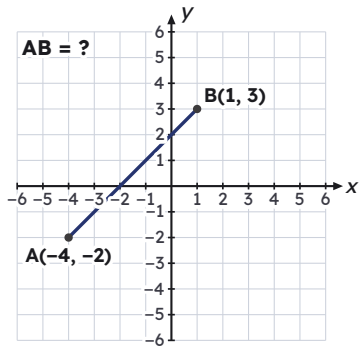
Formula _____

Substitute _____

Answer _____

ROUND 3 Questions 5–6: find the length of AB, correct to one decimal place.

5

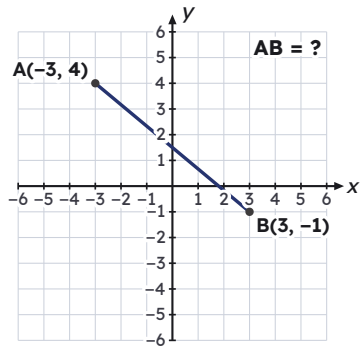


Formula _____

Substitute _____

Answer _____

6



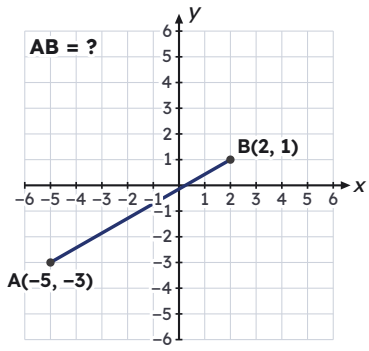
Formula _____

Substitute _____

Answer _____

ROUND 4 Questions 7–8: find the length of AB, correct to one decimal place.

7

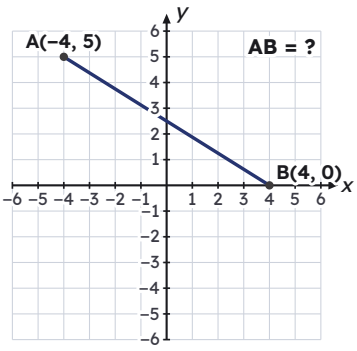


Formula _____

Substitute _____

Answer _____

8



Formula _____

Substitute _____

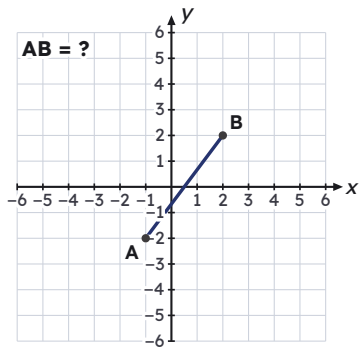
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Read A and B off the grid first: across (x), then up or down (y).

ROUND 5 Questions 9–10: read A and B, then find the length of AB.

9

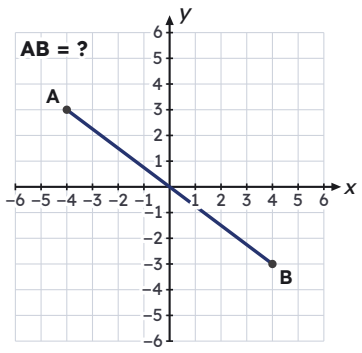


A and B _____

Substitute _____

Answer _____

10



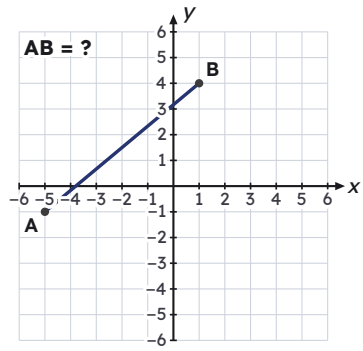
A and B _____

Substitute _____

Answer _____

ROUND 6 Questions 11–12: read A and B, then find the length of AB, correct to one decimal place.

11

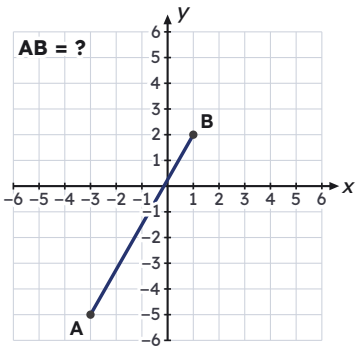


A and B _____

Substitute _____

Answer _____

12



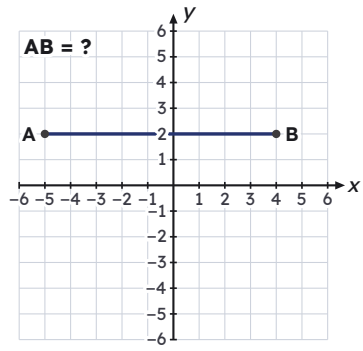
A and B _____

Substitute _____

Answer _____

ROUND 7 Questions 13–14: read A and B, then find the length of AB.

13

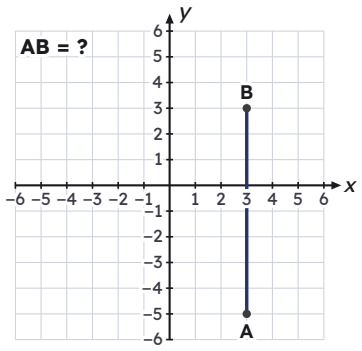


A and B _____

Substitute _____

Answer _____

14



A and B _____

Substitute _____

Answer _____

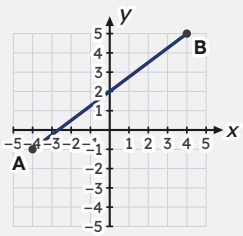
11.1 The length of an interval

MEDIUM

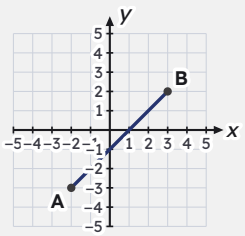
WE DO With your teacher: copy what they write on the board.

Example 2 **Read the points, then find each length. Round to one decimal place where needed.**

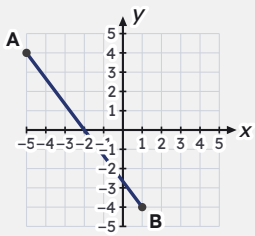
a Length



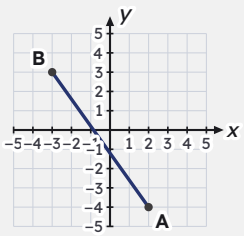
b Length



c Length

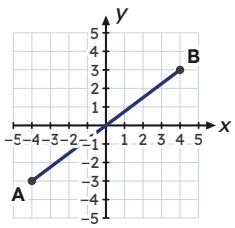


d Length



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

1 Length

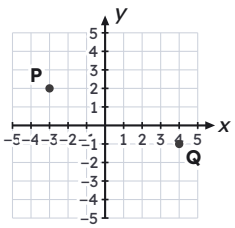


A and B _____

Substitute _____

Answer _____

2 Year 8

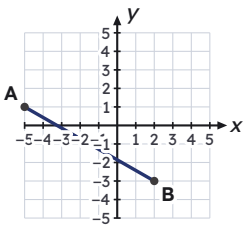


P _____

Q _____

Check _____

3 Length

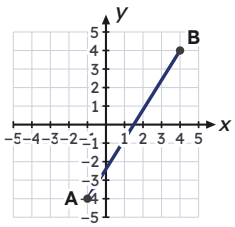


A and B _____

Substitute _____

Answer _____

4 Length

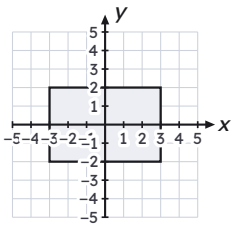


A and B _____

Substitute _____

Answer _____

5 10.4

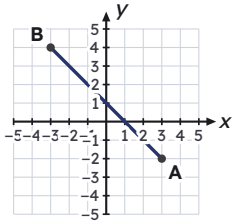


Length _____

Width _____

Area _____

6 Length



A and B _____

Substitute _____

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 On a map, 1 unit = 100 m. A drone flies in a straight line from (−3, 2) to (5, 8). How far does it fly, in metres?

Answer _____

2 Town A is at (2, 1) and town B is at (14, 6) on a map grid in kilometres. The road goes 12 km east, then 5 km north. How much shorter is the straight-line distance than the road?

Answer _____

3 Show that the triangle with vertices A(1, 1), B(7, 1) and C(4, 5) is isosceles, then find its perimeter.

Answer _____

11.2 The midpoint of an interval

EASY

Learning intention We are learning to find the midpoint of an interval on the number plane.

Success criteria
- ☐ I can read the coordinates of the endpoints of an interval
- ☐ I can find the midpoint by averaging the x-values and the y-values
- ☐ I can find midpoints with negative and decimal coordinates

NOTES — Key ideas for this lesson

TEACHER COPY

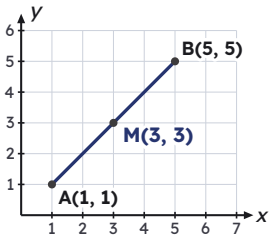
KEY TERMS

midpoint (M): the point halfway along an interval
average: add, then divide by how many
endpoints: A(x_1, y_1) and B(x_2, y_2)

FORMULAS

M = ((x_1 + x_2) ÷ 2, (y_1 + y_2) ÷ 2)
x of M = average of the x-values
y of M = average of the y-values
Endpoint: B = 2 × M − A (for x and y)

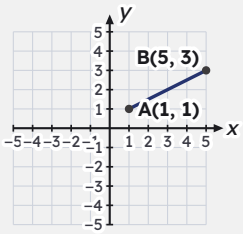
STEPS
1. write the two points
2. add the x-values, ÷ 2
3. add the y-values, ÷ 2
4. write M as a point (x, y)



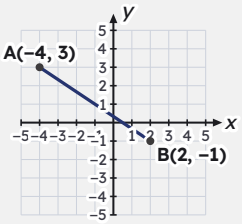
WE DO — With your teacher: copy what they write on the board.

Example 1 Find the midpoint M of each interval AB.

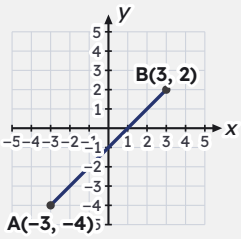
a Midpoint



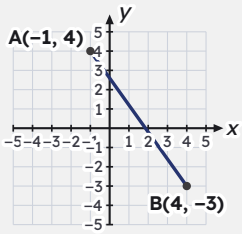
b Midpoint



c Midpoint



d Midpoint



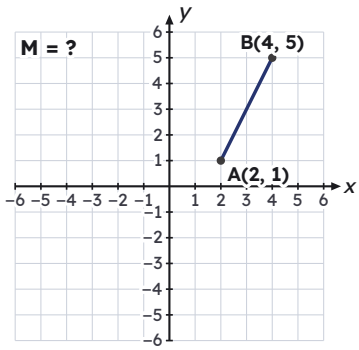
YOU DO — On your own: work down the box, one round at a time. Show your working.

2 QUESTIONS

REMEMBER Add the two x-values and halve · add the two y-values and halve · write the answer as a point

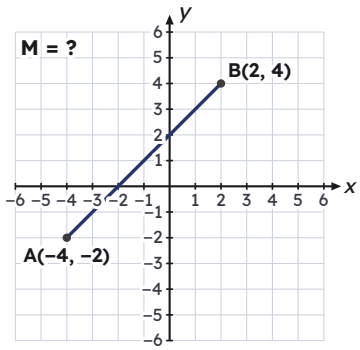
ROUND 1 Questions 1–2: find the midpoint M of AB.

1



Formula M = ((x_1+x_2)÷2, (y_1+y_2)÷2)
Substitute _____
Answer _____

2



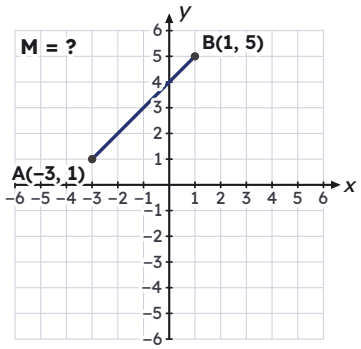
Formula M = ((x_1+x_2)÷2, (y_1+y_2)÷2)
Substitute _____
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Watch the signs: (−5 + 1) ÷ 2 = −2 · a midpoint can be a decimal, like 0.5.

ROUND 2 Questions 3–4: find the midpoint M of AB.

3

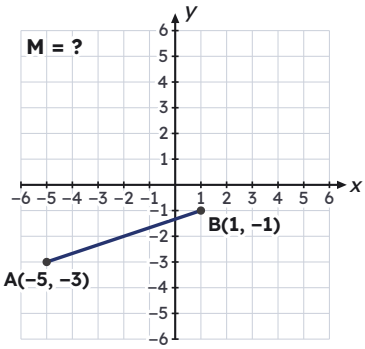


Formula _____

Substitute _____

Answer _____

4



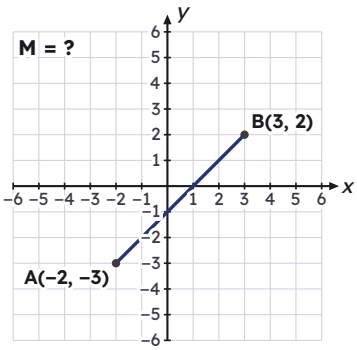
Formula _____

Substitute _____

Answer _____

ROUND 3 Questions 5–6: find the midpoint M of AB. The answers have halves.

5

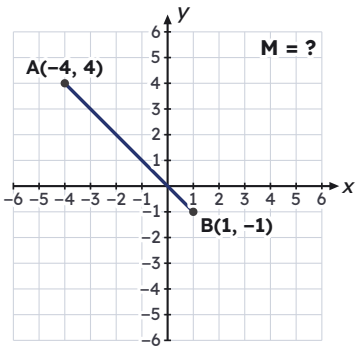


Formula _____

Substitute _____

Answer _____

6



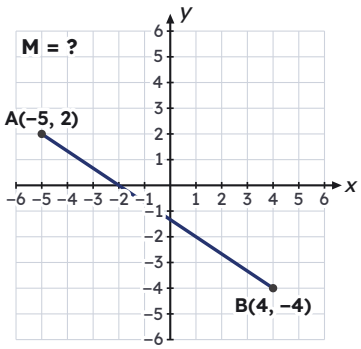
Formula _____

Substitute _____

Answer _____

ROUND 4 Questions 7–8: find the midpoint M of AB.

7

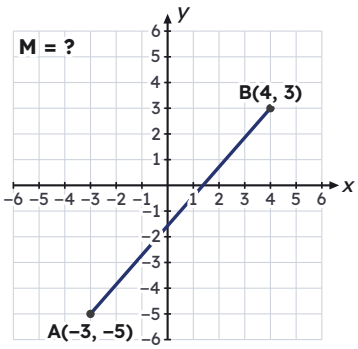


Formula _____

Substitute _____

Answer _____

8



Formula _____

Substitute _____

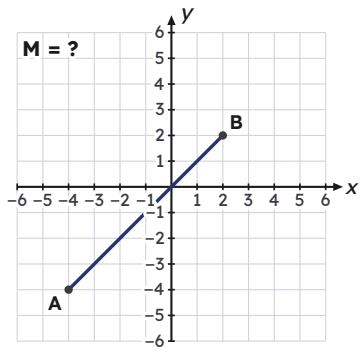
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Read A and B off the grid first: across (x), then up or down (y).

ROUND 5 Questions 9–10: read A and B, then find the midpoint M.

9

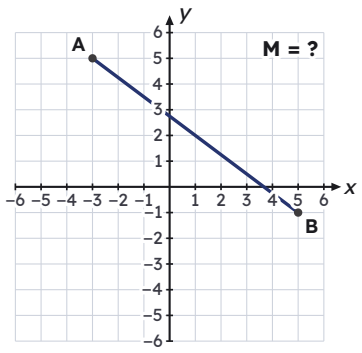


A and B _____

Substitute _____

Answer _____

10



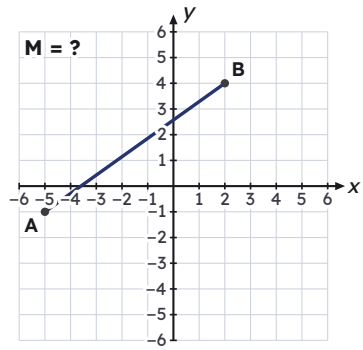
A and B _____

Substitute _____

Answer _____

ROUND 6 Questions 11–12: read A and B, then find the midpoint M.

11

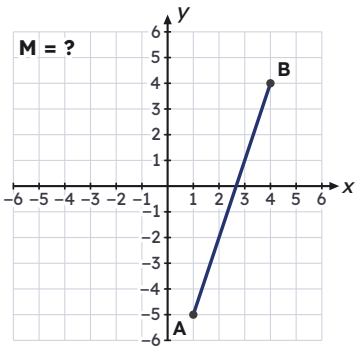


A and B _____

Substitute _____

Answer _____

12



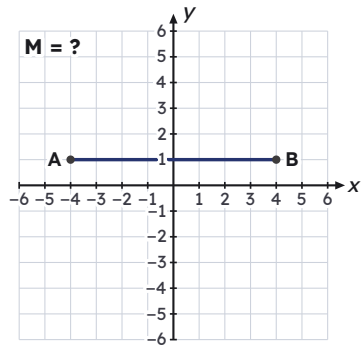
A and B _____

Substitute _____

Answer _____

ROUND 7 Questions 13–14: read A and B, then find the midpoint M.

13

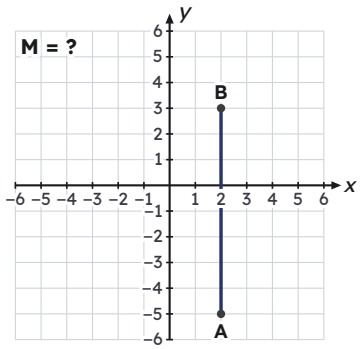


A and B _____

Substitute _____

Answer _____

14



A and B _____

Substitute _____

Answer _____

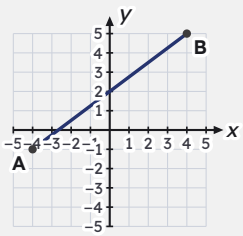
11.2 The midpoint of an interval

MEDIUM

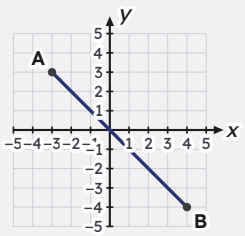
WE DO With your teacher: copy what they write on the board.

Example 2 **Read the points, then find each midpoint (or the endpoint B).**

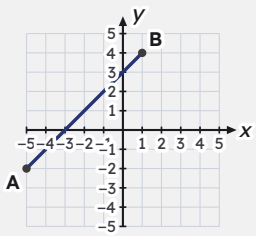
a Midpoint



b Midpoint



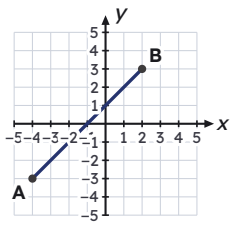
c Midpoint



d A is (−3, 2) and M is (1, −1). Find B.

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

1 Midpoint

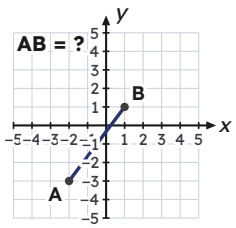


A and B _____

Substitute _____

Answer _____

2 11.1

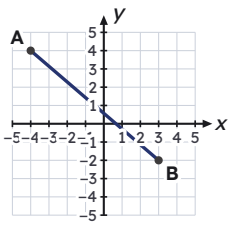


A and B _____

Substitute _____

Answer _____

3 Midpoint

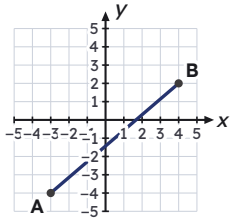


A and B _____

Substitute _____

Answer _____

4 Midpoint

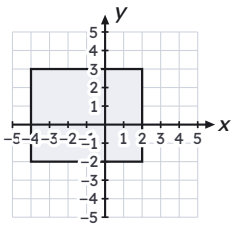


A and B _____

Substitute _____

Answer _____

5 10.4

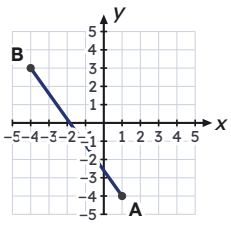


Length _____

Width _____

Perimeter _____

6 Midpoint



A and B _____

Substitute _____

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 Two friends live at (−4, 3) and (8, −1) on a town map (units in km). They meet halfway between their homes. Where do they meet?

Answer _____

2 A circle has a diameter with endpoints (−1, 2) and (7, 8). Find the centre and the radius of the circle.

Answer _____

3 A straight path runs from a gate G(2, 1) to a shed S(10, 7) (units in metres). A tap is halfway along the path. Find the tap's position and how far it is from the gate.

Answer _____

11.3 The gradient of an interval

EASY

Learning intention We are learning to find the gradient (steepness) of an interval on the number plane.

Success criteria
- ☐ I can read the coordinates of the endpoints of an interval
- ☐ I can use the gradient formula with the two points
- ☐ I can say whether a gradient is positive, negative, zero or undefined

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

gradient (m): the steepness: rise ÷ run
rise: change in y: up +, down –
run: change in x, left to right
positive: goes up to the right
negative: goes down to the right
zero: horizontal
undefined: vertical (run = 0)

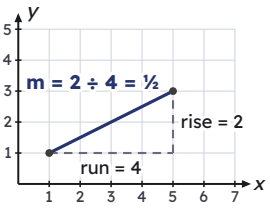
FORMULAS

m = rise ÷ run

m = (y₂ – y₁) ÷ (x₂ – x₁)

Simplify the fraction

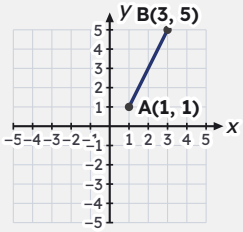
STEPS 1 write the two points
2 rise: $y_2 - y_1$ 3 run: $x_2 - x_1$
4 m = rise ÷ run, simplified



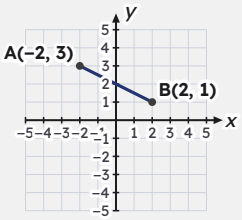
WE DO — With your teacher: copy what they write on the board.

Example 1 Find the gradient of each interval AB.

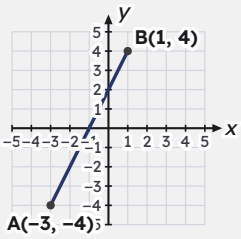
a Gradient



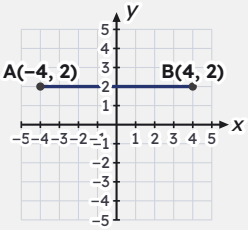
b Gradient



c Gradient



d Gradient



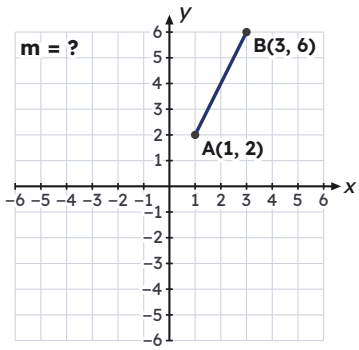
YOU DO — On your own: work down the box, one round at a time. Show your working.

2 QUESTIONS

REMEMBER m = rise ÷ run · subtract the y-values, then the x-values, in the same order · simplify

ROUND 1 — Questions 1–2: find the gradient of AB.

1

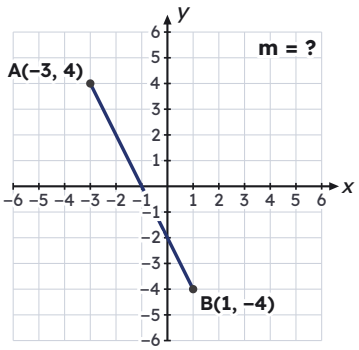


Formula m = (y₂–y₁) ÷ (x₂–x₁)

Substitute _____

Answer _____

2



Formula m = (y₂–y₁) ÷ (x₂–x₁)

Substitute _____

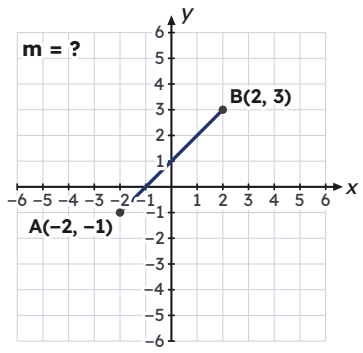
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Up to the right: positive · down to the right: negative · a negative ÷ a positive is negative.

ROUND 2 Questions 3–4: find the gradient of AB.

3

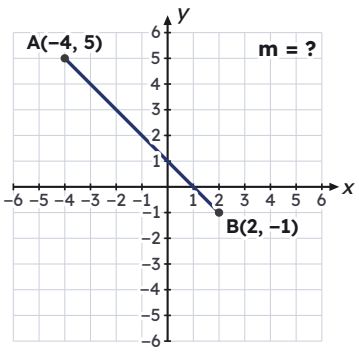


Formula _____

Substitute _____

Answer _____

4



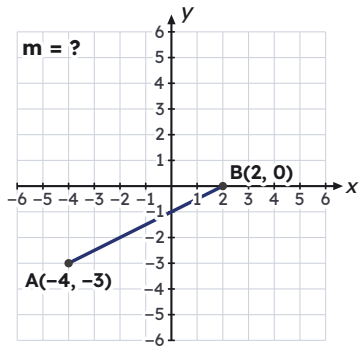
Formula _____

Substitute _____

Answer _____

ROUND 3 Questions 5–6: find the gradient of AB. Simplify the fraction.

5

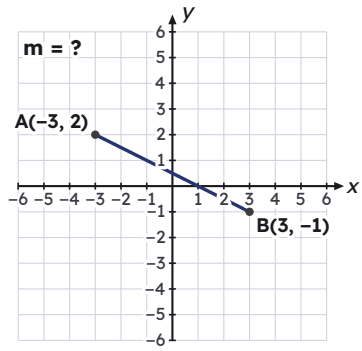


Formula _____

Substitute _____

Answer _____

6



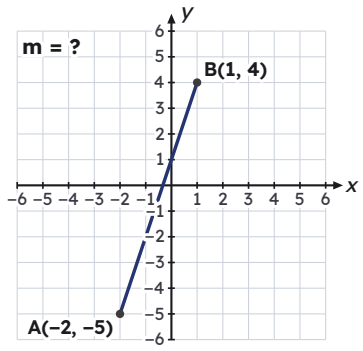
Formula _____

Substitute _____

Answer _____

ROUND 4 Questions 7–8: find the gradient of AB.

7

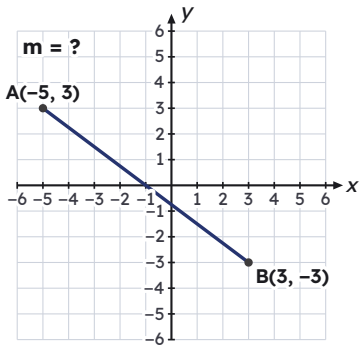


Formula _____

Substitute _____

Answer _____

8



Formula _____

Substitute _____

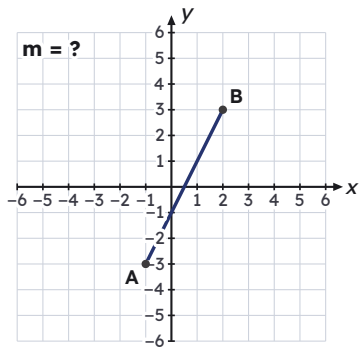
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Read A and B off the grid first · horizontal: m = 0 · vertical: m is undefined.

ROUND 5 Questions 9–10: read A and B, then find the gradient of AB.

9

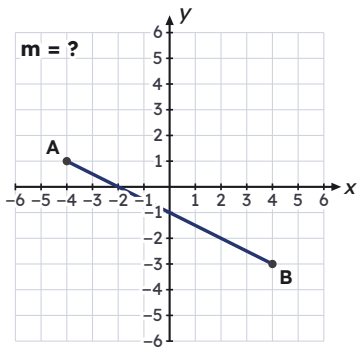


A and B _____

Substitute _____

Answer _____

10



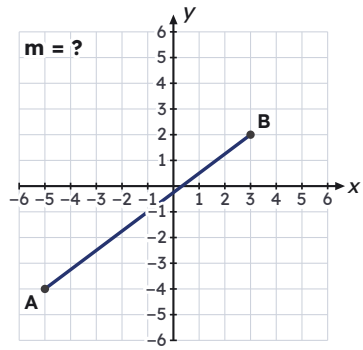
A and B _____

Substitute _____

Answer _____

ROUND 6 Questions 11–12: read A and B, then find the gradient of AB.

11

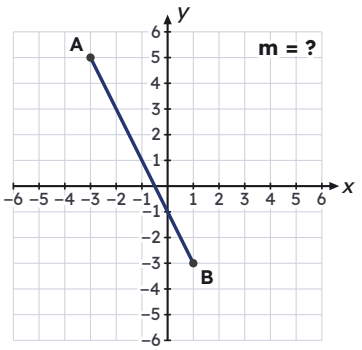


A and B _____

Substitute _____

Answer _____

12



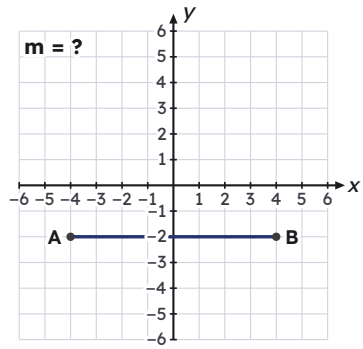
A and B _____

Substitute _____

Answer _____

ROUND 7 Questions 13–14: read A and B. Is the gradient zero or undefined?

13

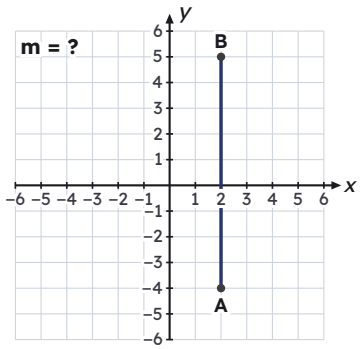


A and B _____

Substitute _____

Answer _____

14



A and B _____

Substitute _____

Answer _____

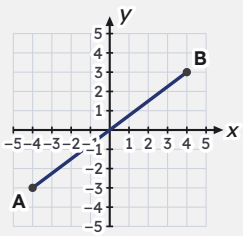
11.3 The gradient of an interval

MEDIUM

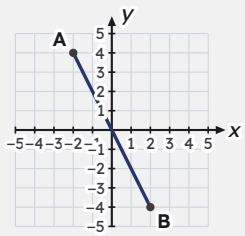
WE DO With your teacher: copy what they write on the board.

Example 2 **Read the points, then find each gradient.**

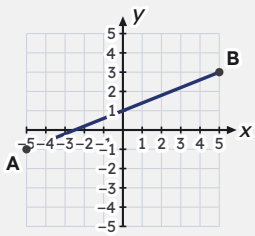
a Gradient



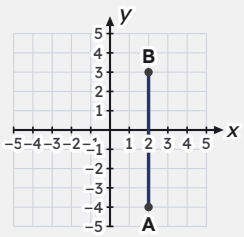
b Gradient



c Gradient

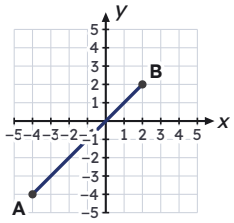


d Gradient



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

1 Gradient

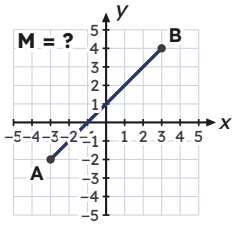


A and B _____

Substitute _____

Answer _____

2 11.2

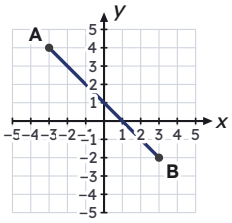


A and B _____

Substitute _____

Answer _____

3 Gradient

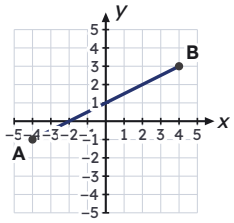


A and B _____

Substitute _____

Answer _____

4 Gradient

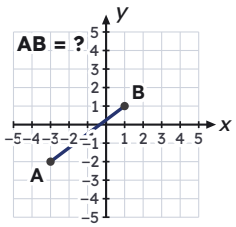


A and B _____

Substitute _____

Answer _____

5 11.1

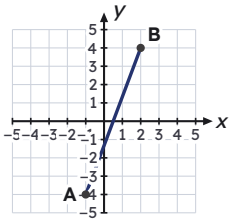


A and B _____

Substitute _____

Answer _____

6 Gradient



A and B _____

Substitute _____

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A road climbs 45 m over a horizontal distance of 600 m. Find its gradient as a decimal and as a percentage.

Answer _____

2 A roof rises 2.4 m over a horizontal distance of 4 m. A builder says the roof is steeper than a ramp with gradient ½. Is the builder right? Explain.

Answer _____

3 Show that the points A(0, 1), B(2, 5) and C(5, 11) lie on one straight line.

Answer _____

11.4 Graphing linear equations

EASY

Learning intention We are learning to graph straight lines from a table of values and find where they cross the axes.

Success criteria
- ☐ I can complete a table of values
- ☐ I can plot the points and rule a straight line through them
- ☐ I can find the x- and y-intercepts of a line

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

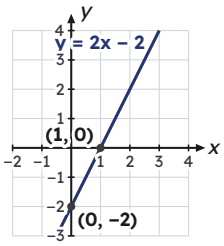
linear equation: its graph is a straight line
table of values: x-values and the y-values they give
x-intercept: where a line crosses the x-axis (y = 0)
y-intercept: where a line crosses the y-axis (x = 0)
x = a: a vertical line
y = b: a horizontal line

FORMULAS

x-intercept: put y = 0 and solve
y-intercept: put x = 0
On the line? Substitute x and y: both sides equal

STEPS

1. substitute each x into the rule
2. write the y-values in the table
3. plot the points
4. rule a line through them



WE DO — With your teacher: copy what they write on the board.

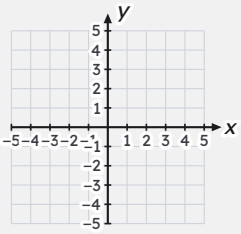
Example 1 For y = 2x + 1:

a Table

y = 2x + 1

x	−2	−1	0	1	2
y					

b Graph



c Find the x- and y-intercepts.

d Does (3, 7) lie on the line?

YOU DO — On your own: work down the box, one round at a time. Show your working.

2 QUESTIONS

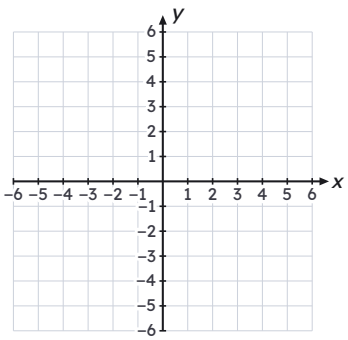
REMEMBER Substitute each x into the rule · brackets for negatives: 2 × (−2) + 1 = −3 · plot every point, then rule the line

ROUND 1 — Questions 1–2: complete the table, graph the line, then give both intercepts.

1

y = x + 2

x	−2	−1	0	1	2
y					



y-values _____

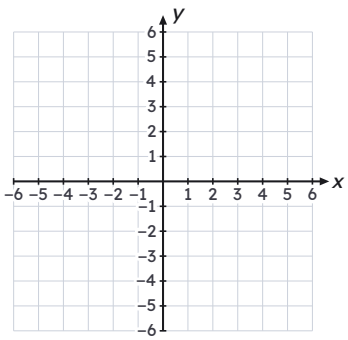
x-intercept _____

y-intercept _____

2

y = −x + 3

x	−2	−1	0	1	2
y					



y-values _____

x-intercept _____

y-intercept _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

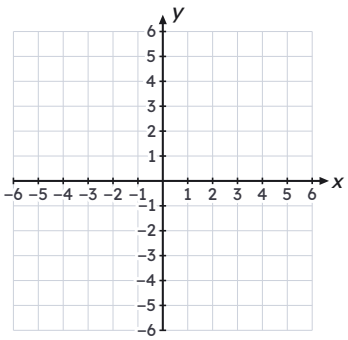
REMEMBER x-intercept: where the line crosses the x-axis (y = 0) · y-intercept: where it crosses the y-axis (x = 0)

ROUND 2 Questions 3–4: complete the table, graph the line, then give both intercepts.

3

$y = 2x - 2$

x	−2	−1	0	1	2
y					



y-values _____

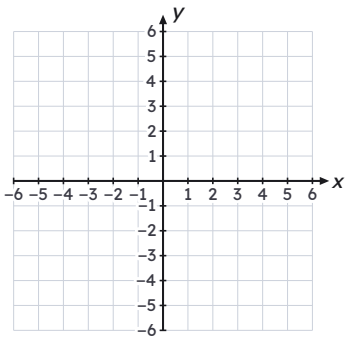
x-intercept _____

y-intercept _____

4

$y = x - 2$

x	−2	−1	0	1	2
y					



y-values _____

x-intercept _____

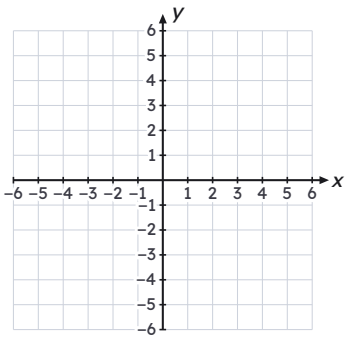
y-intercept _____

ROUND 3 Questions 5–6: complete the table, graph the line, then give both intercepts.

5

$y = -2x + 2$

x	−2	−1	0	1	2
y					



y-values _____

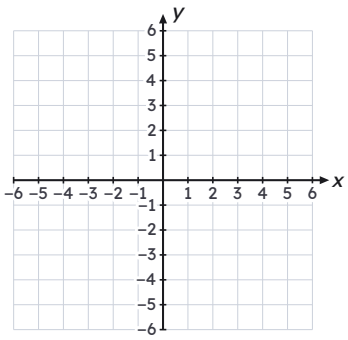
x-intercept _____

y-intercept _____

6

$y = 3x$

x	−2	−1	0	1	2
y					



y-values _____

x-intercept _____

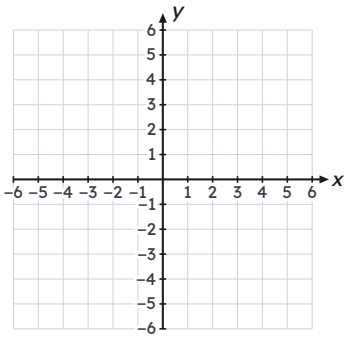
y-intercept _____

ROUND 4 Questions 7–8: complete the table, graph the line, then give both intercepts.

7

$y = x + 3$

x	−2	−1	0	1	2
y					



y-values _____

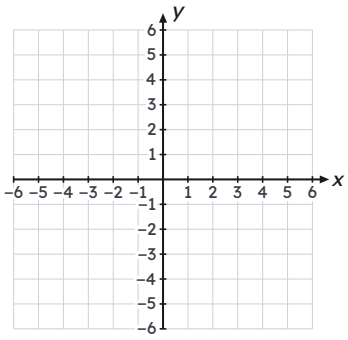
x-intercept _____

y-intercept _____

8

$y = -x - 2$

x	−2	−1	0	1	2
y					



y-values _____

x-intercept _____

y-intercept _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

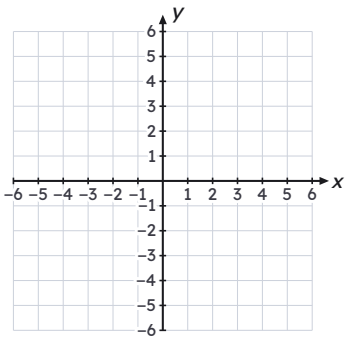
REMEMBER A fraction rule: ½ × (−2) = −1 · a negative gradient: the line goes down to the right.

ROUND 5 Questions 9–10: complete the table, graph the line, then give both intercepts.

9

$y = \frac{1}{2}x + 1$

x	−2	−1	0	1	2
y					



y-values _____

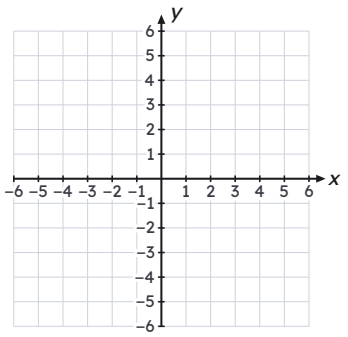
x-intercept _____

y-intercept _____

10

$y = -\frac{1}{2}x + 2$

x	−2	−1	0	1	2
y					



y-values _____

x-intercept _____

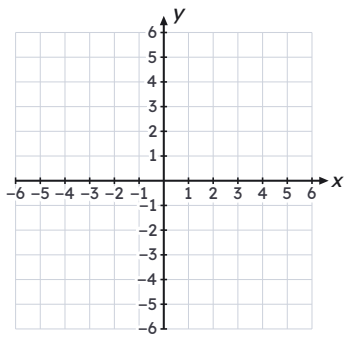
y-intercept _____

ROUND 6 Questions 11–12: complete the table, graph the line, then give both intercepts.

11

$y = 2x - 4$

x	−2	−1	0	1	2
y					



y-values _____

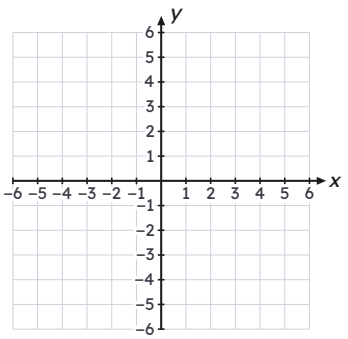
x-intercept _____

y-intercept _____

12

$y = -3x$

x	−2	−1	0	1	2
y					



y-values _____

x-intercept _____

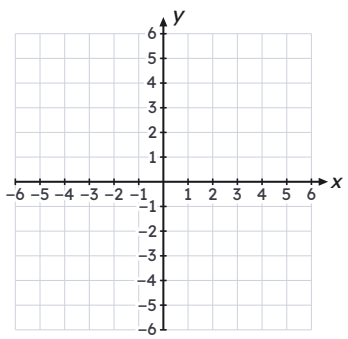
y-intercept _____

ROUND 7 Questions 13–14: complete the table, graph the line, then give both intercepts.

13

$y = -2x - 2$

x	−2	−1	0	1	2
y					



y-values _____

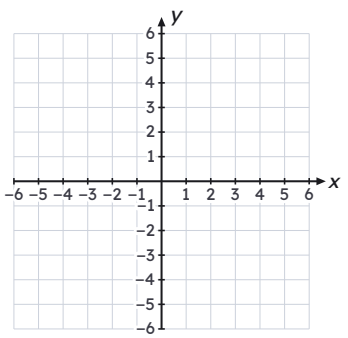
x-intercept _____

y-intercept _____

14

$y = x - 3$

x	−2	−1	0	1	2
y					



y-values _____

x-intercept _____

y-intercept _____

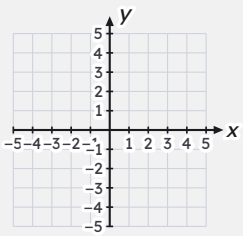
11.4 Graphing linear equations

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Graph, test or find as shown.**

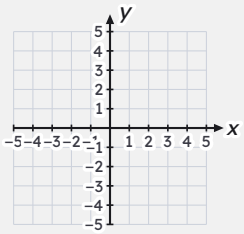
a Graph $y = -2x + 3$



b Find the x- and y-intercepts of $y = -2x + 3$.

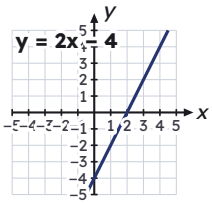
c Find the x- and y-intercepts of $3x + 2y = 6$.

d Graph $x = -2$ and $y = 3$



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

1 Intercepts

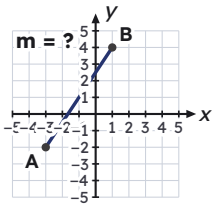


x-intercept _____

y-intercept _____

Check _____

2 11.3

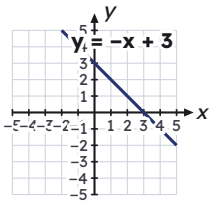


A and B _____

Substitute _____

Answer _____

3 Intercepts

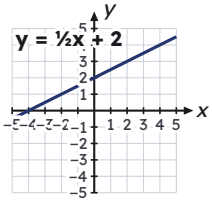


x-intercept _____

y-intercept _____

Check _____

4 Intercepts

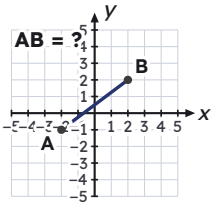


x-intercept _____

y-intercept _____

Check _____

5 11.1

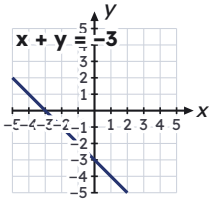


A and B _____

Substitute _____

Answer _____

6 Intercepts



x-intercept _____

y-intercept _____

Check _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A plumber charges $C = 40 + 60h$ dollars for h hours of work. Make a table for $h = 0$, 1, 2 and 3. What is the C-intercept, and what does it mean?

Answer _____

2 The line $3x + 2y = 18$ crosses both axes. Find where, then find the length of the interval between these two points, correct to one decimal place.

Answer _____

3 The lines $x = -1$, $x = 5$, $y = 2$ and $y = -3$ enclose a rectangle. Find its perimeter and its area.

Answer _____

11.5 The gradient–intercept equation

EASY

Learning intention We are learning to use y = mx + c to find the gradient and y-intercept of a line, and to graph it.

Success criteria
- ☐ I can state the gradient and y-intercept of a line in the form y = mx + c
- ☐ I can rearrange an equation into the form y = mx + c
- ☐ I can graph a line from its gradient and y-intercept

NOTES — Key ideas for this lesson

TEACHER COPY

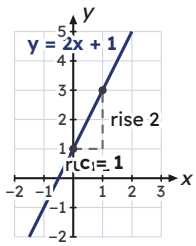
KEY TERMS

y = mx + c: the gradient–intercept form
m: the gradient: the number in front of x
c: the y-intercept: the number on its own
parallel lines: lines with the same gradient
subject: the variable on its own on one side (y = …)

FORMULAS

y = mx + c
m = gradient, c = y-intercept
Parallel lines: equal gradients

STEPS 1 make y the subject
2 read m and c 3 plot (0, c)
4 go across 1 and up m; rule the line



WE DO — With your teacher: copy what they write on the board.

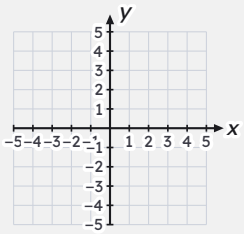
Example 1 State m and c, then graph the line.

a y = 3x + 1

b y = 8 − x

c y = ⅕x + 4

d Graph y = 2x − 3



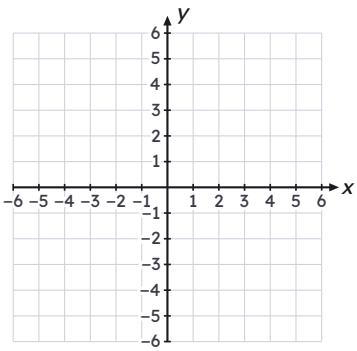
YOU DO — On your own: work down the box, one round at a time. Show your working.

2 QUESTIONS

REMEMBER m is the number in front of x (with its sign) · c is the number on its own · start at (0, c), then across 1 and up m

ROUND 1 Questions 1–2: state m and c, then graph the line.

1 y = 2x + 1

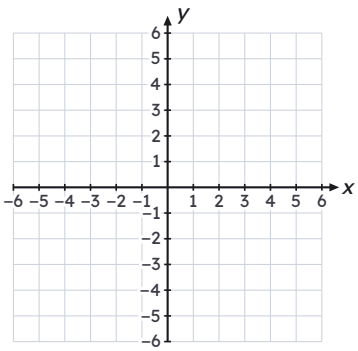


Formula y = mx + c

m and c

Two points

2 y = −x + 4



Formula y = mx + c

m and c

Two points

11.5 The gradient–intercept equation

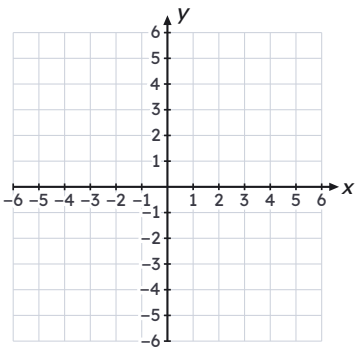
EASY

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER A negative m: go across 1 and down · m = ½: go across 2 and up 1.

ROUND 2 Questions 3–4: state m and c, then graph the line.

3 y = 3x − 2

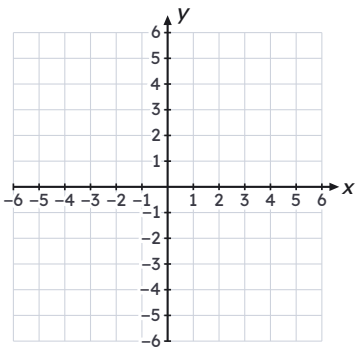


Formula _____

m and c _____

Two points _____

4 y = −2x + 3



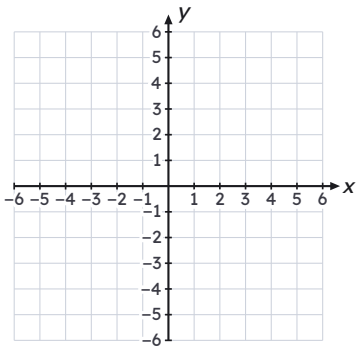
Formula _____

m and c _____

Two points _____

ROUND 3 Questions 5–6: state m and c, then graph the line.

5 y = x − 4

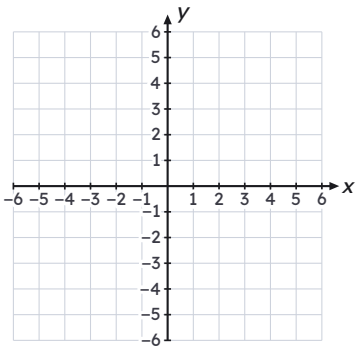


Formula _____

m and c _____

Two points _____

6 y = ½x + 2



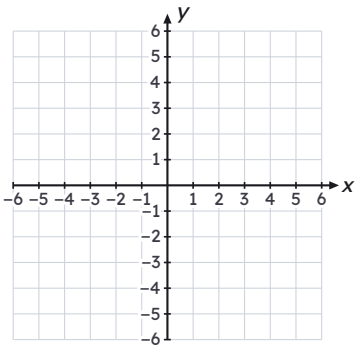
Formula _____

m and c _____

Two points _____

ROUND 4 Questions 7–8: state m and c, then graph the line.

7 y = −x − 1

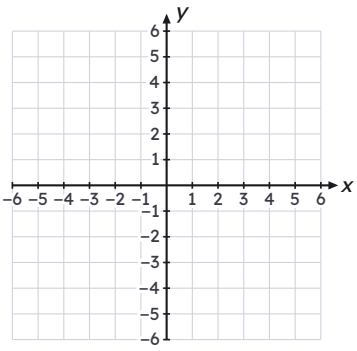


Formula _____

m and c _____

Two points _____

8 y = 2x



Formula _____

m and c _____

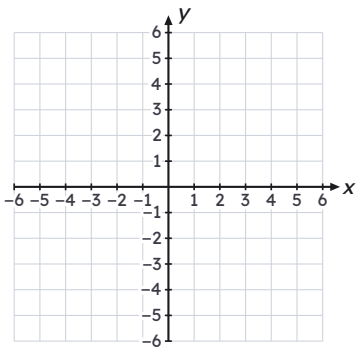
Two points _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Get y on its own first: move the x-term, then divide every term.

ROUND 5 Questions 9–10: rearrange into y = mx + c, then graph the line.

9 $2x + y = 4$

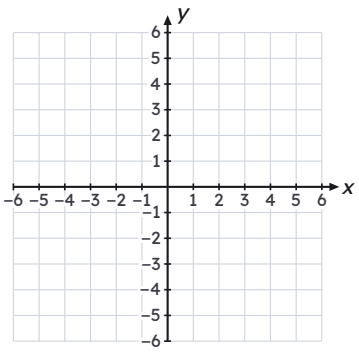


Rearrange _____

m and c _____

Two points _____

10 $y - 3x = 1$



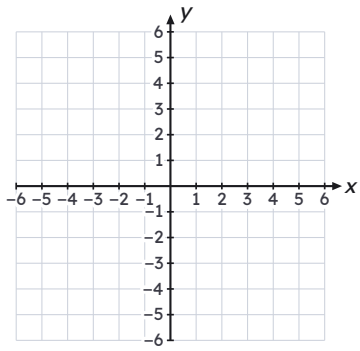
Rearrange _____

m and c _____

Two points _____

ROUND 6 Questions 11–12: rearrange into y = mx + c, then graph the line.

11 $x + y = -2$

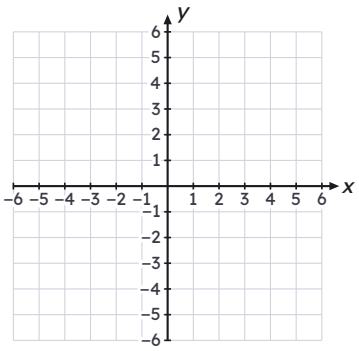


Rearrange _____

m and c _____

Two points _____

12 $2y = x + 4$



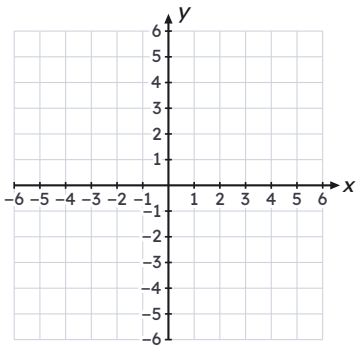
Rearrange _____

m and c _____

Two points _____

ROUND 7 Questions 13–14: rearrange into y = mx + c, then graph the line.

13 $4x + 2y = 6$

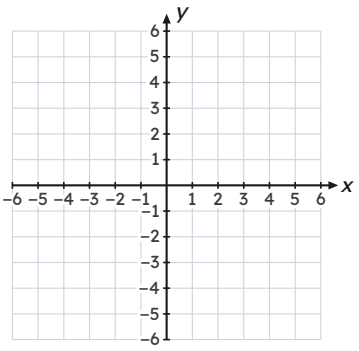


Rearrange _____

m and c _____

Two points _____

14 $y + 2 = x$



Rearrange _____

m and c _____

Two points _____

11.5 The gradient–intercept equation

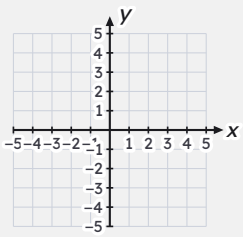
MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Find, graph or write as shown.**

a State m and c for $3x + 2y = 12$.

b Graph $y = -2x + 4$

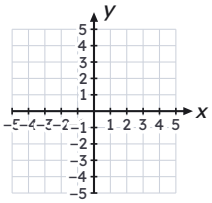


c Write the equation with gradient -3 and y-intercept 5.

d Write the equation of the line parallel to $y = 4x - 1$ with y-intercept 3.

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

1 $y = -3x + 2$

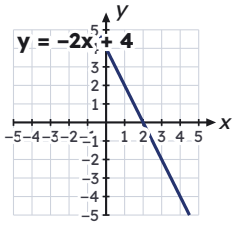


Rearrange _____

m and c _____

Two points _____

2 11.4

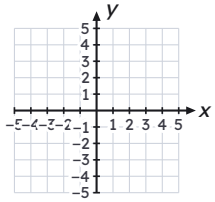


x-intercept _____

y-intercept _____

Check _____

3 $2x + y = -1$

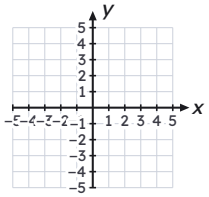


Rearrange _____

m and c _____

Two points _____

4 $y = -\frac{1}{2}x + 1$

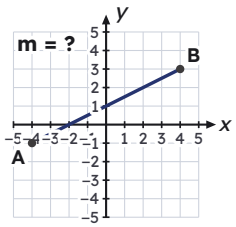


Rearrange _____

m and c _____

Two points _____

5 11.3

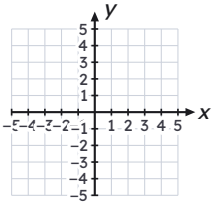


A and B _____

Substitute _____

Answer _____

6 $3x - y = 2$



Rearrange _____

m and c _____

Two points _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A phone plan costs $C = 15 + 0.5g$ dollars a month for g gigabytes of data. State the gradient and the C-intercept, and say what each means.

Answer _____

2 Line A has equation $y = 2x + 3$. Line B is parallel to line A and crosses the y-axis 5 units below it. Write the equation of line B, then find its x-intercept.

Answer _____

3 A student rewrote $2y - 6x = 8$ as $y = 6x + 8$. Find the mistake, write the correct equation and state m and c.

Answer _____

11.6 Finding the equation of a line

EASY

Learning intention We are learning to find the equation of a straight line from its graph.

Success criteria
- ☐ I can read the y-intercept where the line crosses the y-axis
- ☐ I can find the gradient from two points on the line
- ☐ I can write the equation of the line as y = mx + c

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

y-intercept (c): where the line crosses the y-axis
gradient (m): rise ÷ run between two points
y = mx + c: the equation of the line

FORMULAS

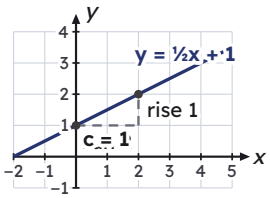
$y = mx + c$

$m = (y_2 - y_1) \div (x_2 - x_1)$

Two points not on the y-axis: find m, then substitute a point to find c

STEPS

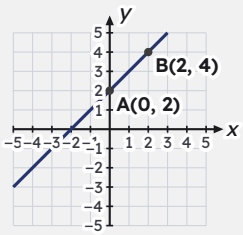
1. read c where the line crosses the y-axis
2. find m from A and B
3. write y = mx + c
4. check with another point



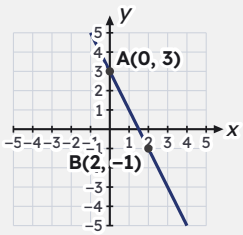
WE DO — With your teacher: copy what they write on the board.

Example 1 Find the equation of each line.

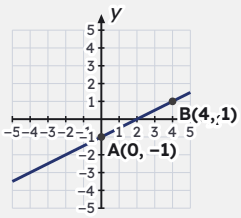
a Equation



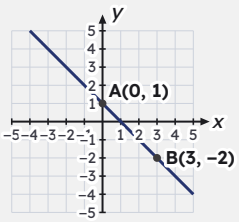
b Equation



c Equation



d Equation



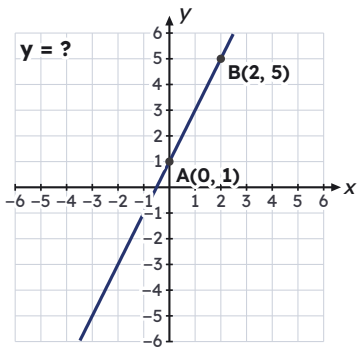
YOU DO — On your own: work down the box, one round at a time. Show your working.

2 QUESTIONS

REMEMBER c: where the line crosses the y-axis · m = rise ÷ run from A to B · then write y = mx + c

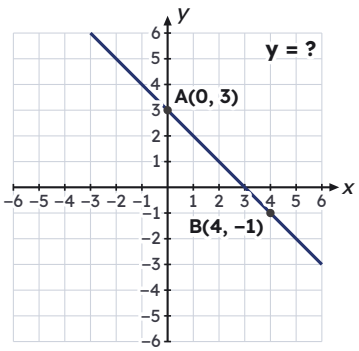
ROUND 1 Questions 1–2: find the equation of the line.

1



Formula: y = mx + c
m and c: _____
Equation: _____

2



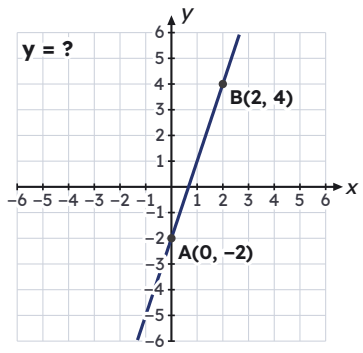
Formula: y = mx + c
m and c: _____
Equation: _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Check your equation: substitute B's x-value and you should get B's y-value.

ROUND 2 Questions 3–4: find the equation of the line.

3

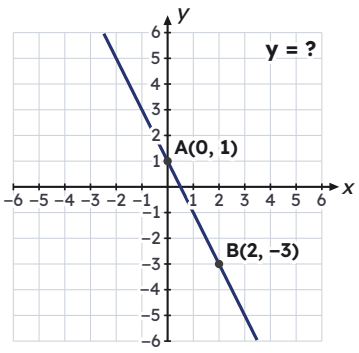


Formula _____

m and c _____

Equation _____

4



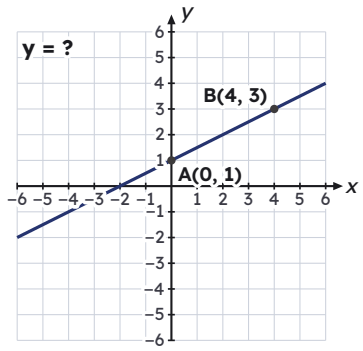
Formula _____

m and c _____

Equation _____

ROUND 3 Questions 5–6: find the equation of the line.

5

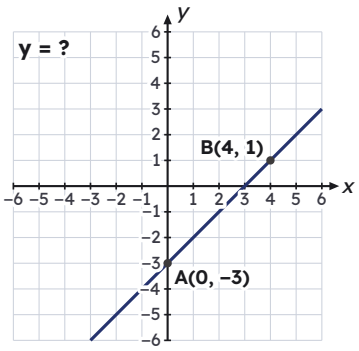


Formula _____

m and c _____

Equation _____

6



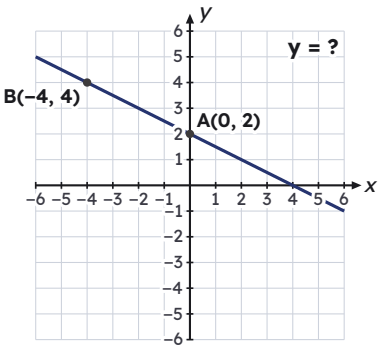
Formula _____

m and c _____

Equation _____

ROUND 4 Questions 7–8: find the equation of the line.

7

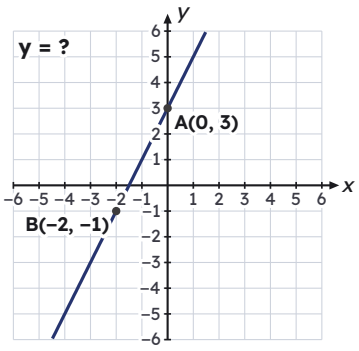


Formula _____

m and c _____

Equation _____

8



Formula _____

m and c _____

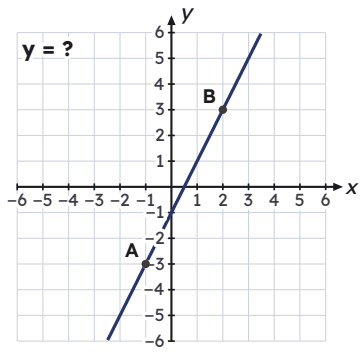
Equation _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Read A and B off the grid · read c where the line crosses the y-axis.

ROUND 5 Questions 9–10: read A and B, then find the equation of the line.

9

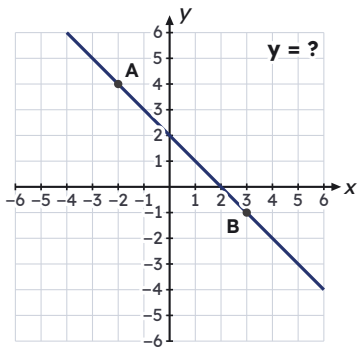


A and B _____

m and c _____

Equation _____

10



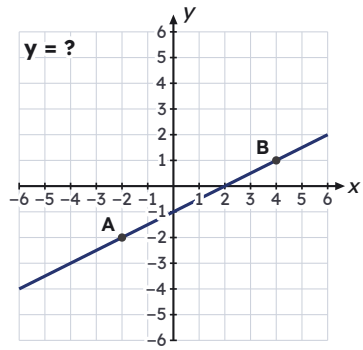
A and B _____

m and c _____

Equation _____

ROUND 6 Questions 11–12: read A and B, then find the equation of the line.

11

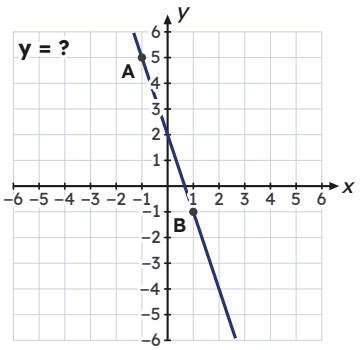


A and B _____

m and c _____

Equation _____

12



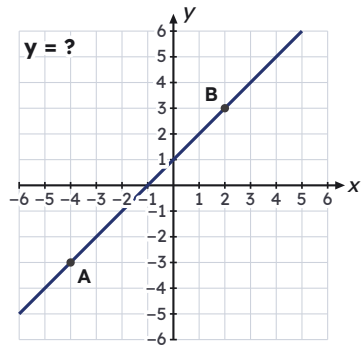
A and B _____

m and c _____

Equation _____

ROUND 7 Questions 13–14: read A and B, then find the equation of the line.

13

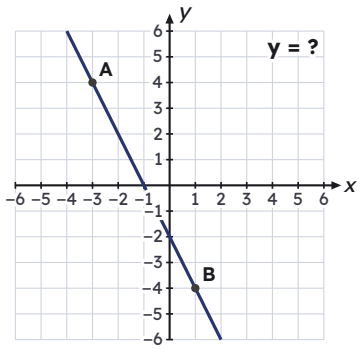


A and B _____

m and c _____

Equation _____

14



A and B _____

m and c _____

Equation _____

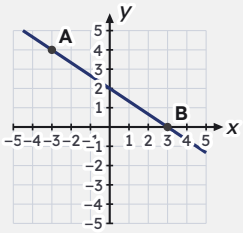
11.6 Finding the equation of a line

MEDIUM

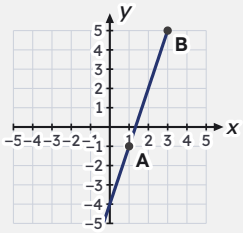
WE DO With your teacher: copy what they write on the board.

Example 2 Read the points, then find the equation of each line.

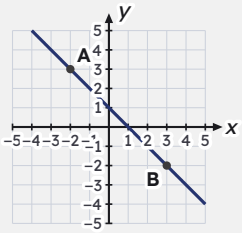
a Equation



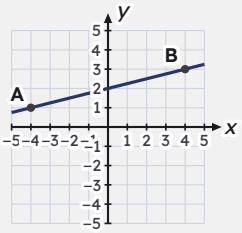
b Equation



c Equation

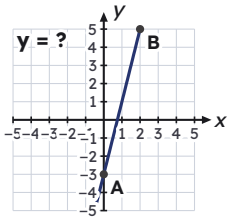


d Equation



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

1 Equation

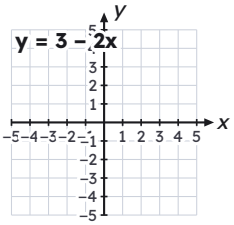


A and B _____

m and c _____

Equation _____

2 11.5

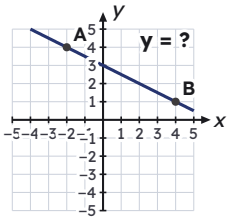


m and c _____

Two points _____

Graph _____

3 Equation

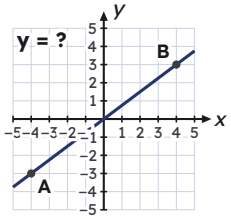


A and B _____

m and c _____

Equation _____

4 Equation

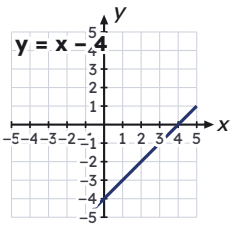


A and B _____

m and c _____

Equation _____

5 11.4

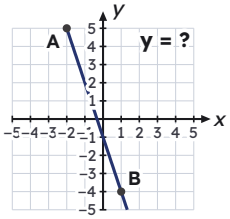


x-intercept _____

y-intercept _____

Check _____

6 Equation



A and B _____

m and c _____

Equation _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A candle is 24 cm tall when it is lit, and 18 cm tall after 2 hours. It burns at a steady rate. Write an equation for its height h cm after t hours, and find when it burns out.

Answer _____

2 A 4 km taxi trip costs $13 and a 10 km trip costs $25. Find an equation for the cost C dollars of a d km trip. What is the flagfall (the charge for 0 km)?

Answer _____

3 Find the equation of the line through A(−2, −1) and B(4, 11), then find where it crosses the x-axis.

Answer _____

11.7 Solving linear equations graphically

EASY

Learning intention We are learning to solve linear equations by reading a straight-line graph.

Success criteria
- ☐ I can use the graph of y = mx + c to solve mx + c = k
- ☐ I can draw the line y = k and read x where the lines meet
- ☐ I can check a graphical solution by substituting

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

solution: the value of x that makes the equation true
point of intersection: where two lines cross
y = k: a horizontal line at height k

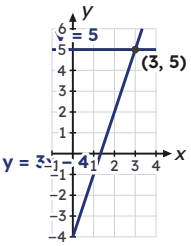
FORMULAS

Solve 3x − 4 = 5: graph y = 3x − 4 and y = 5; read x where they meet

Two lines: the solution is the crossing point (x, y)

Check: substitute x into the equation

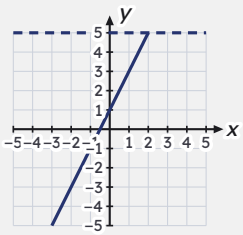
STEPS
1. graph y = mx + c
2. draw the line y = k
3. read x where they meet
4. check by substituting



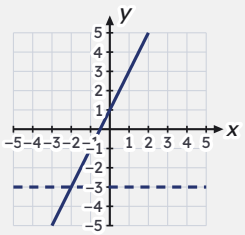
WE DO — With your teacher: copy what they write on the board.

Example 1 Use each graph to solve the equation.

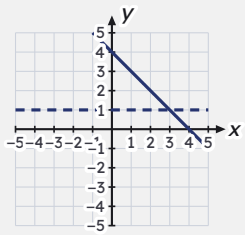
a 2x + 1 = 5



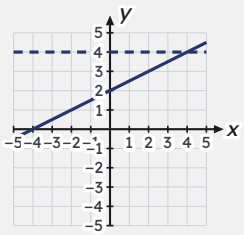
b 2x + 1 = −3



c −x + 4 = 1



d ½x + 2 = 4



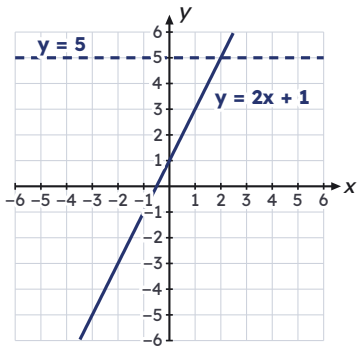
YOU DO — On your own: work down the box, one round at a time. Show your working.

2 QUESTIONS

REMEMBER Find where the dashed line meets the other line · read x straight down (or up) to the x-axis · check by substituting

ROUND 1 — Questions 1–2: read x where the lines meet, then check.

1

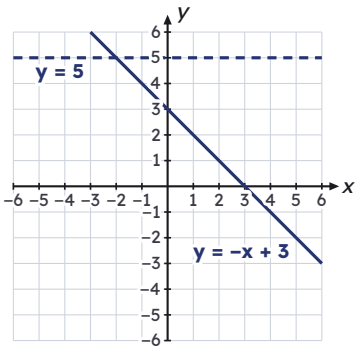


Read x _____

Check _____

Answer _____

2



Read x _____

Check _____

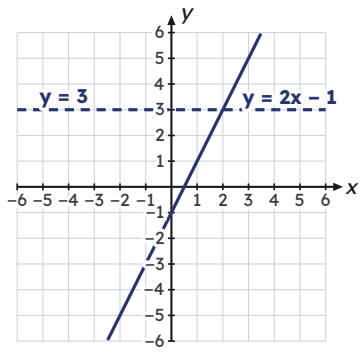
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER The answer is the x-value of the meeting point, not the y-value.

ROUND 2 Questions 3–4: read x where the lines meet, then check.

3

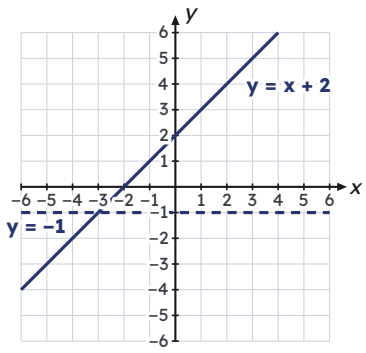


Read x _____

Check _____

Answer _____

4



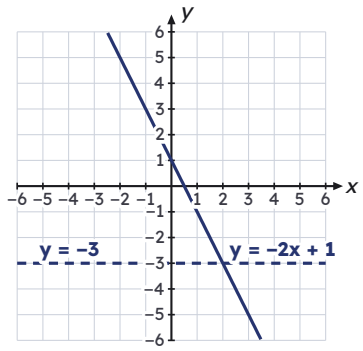
Read x _____

Check _____

Answer _____

ROUND 3 Questions 5–6: read x where the lines meet, then check.

5

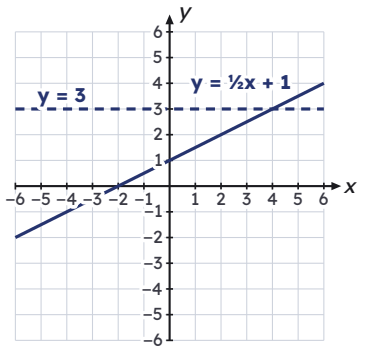


Read x _____

Check _____

Answer _____

6



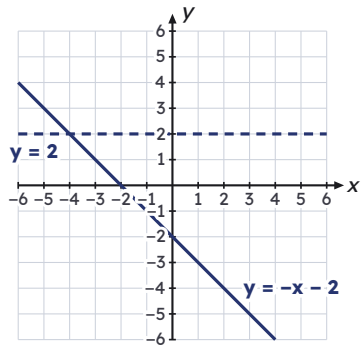
Read x _____

Check _____

Answer _____

ROUND 4 Questions 7–8: read x where the lines meet, then check.

7

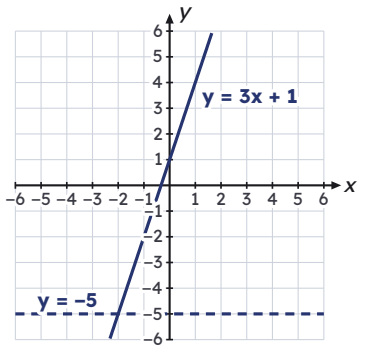


Read x _____

Check _____

Answer _____

8



Read x _____

Check _____

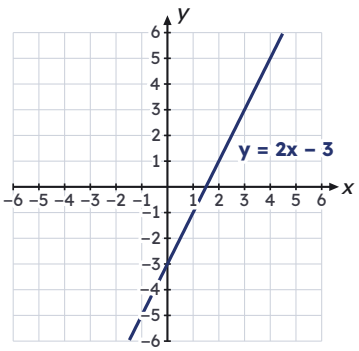
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER To solve 2x − 3 = 1, draw the dashed line y = 1 across the plane first.

ROUND 5 Questions 9–10: draw the line y = k, then solve.

9 2x − 3 = 1

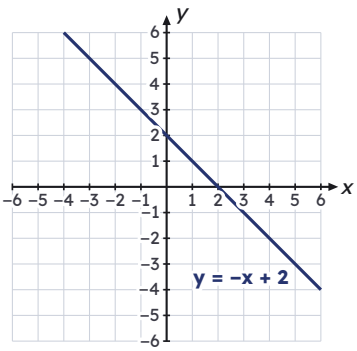


Draw y = _____

Read x _____

Check _____

10 −x + 2 = 5



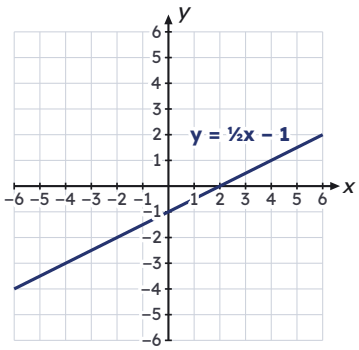
Draw y = _____

Read x _____

Check _____

ROUND 6 Questions 11–12: draw the line y = k, then solve.

11 ½x − 1 = 1

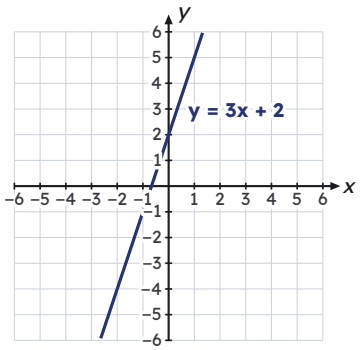


Draw y = _____

Read x _____

Check _____

12 3x + 2 = −4



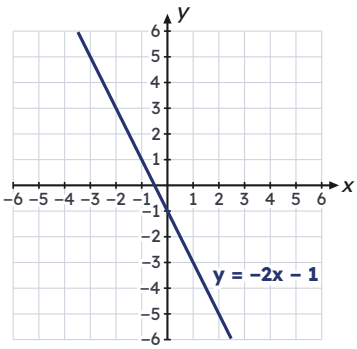
Draw y = _____

Read x _____

Check _____

ROUND 7 Questions 13–14: draw the line y = k, then solve.

13 −2x − 1 = 3

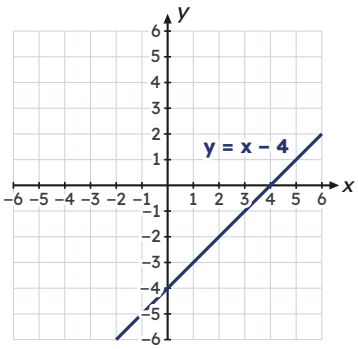


Draw y = _____

Read x _____

Check _____

14 x − 4 = 0



Draw y = _____

Read x _____

Check _____

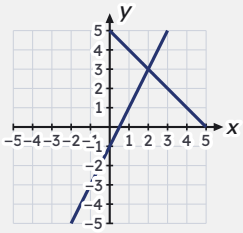
11.7 Solving linear equations graphically

MEDIUM

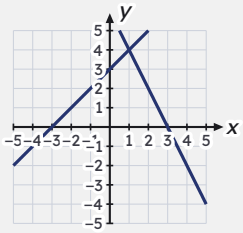
WE DO With your teacher: copy what they write on the board.

Example 2 Find where each pair of lines crosses.

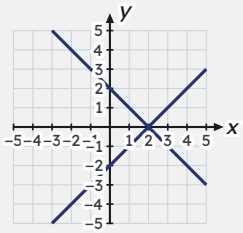
a Crossing point



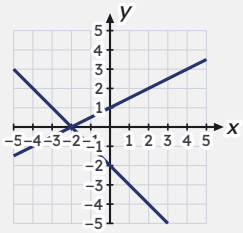
b Crossing point



c Crossing point

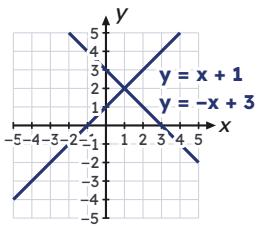


d Crossing point



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

1

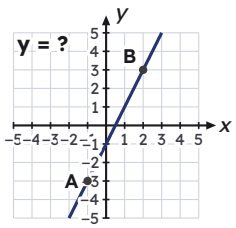


Read x _____

Read y _____

Point _____

2 11.6

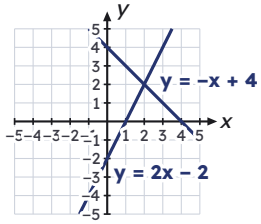


A and B _____

m and c _____

Equation _____

3

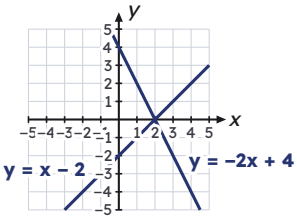


Read x _____

Read y _____

Point _____

4

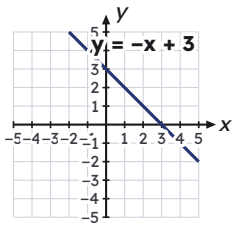


Read x _____

Read y _____

Point _____

5 11.4

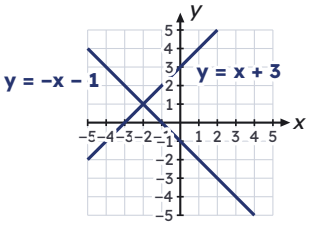


x-intercept _____

y-intercept _____

Check _____

6



Read x _____

Read y _____

Point _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 Plan A costs C = 20 + 5n dollars and plan B costs C = 10 + 7n dollars for n lessons. Graph both for n = 0 to 8. After how many lessons do they cost the same, and how much is it?

Answer _____

2 A tank holds 40 L and drains at 5 L per minute, so V = 40 − 5t. Graph V for t = 0 to 8 and use it to find when the tank holds 15 L.

Answer _____

3 Sam has $30 and saves $12 a week. Mia has $90 and spends $8 a week. Graph both amounts. After how many weeks do they have the same amount, and how much is it?

Answer _____

11.8 Linear modelling

EASY

Learning intention We are learning to use straight-line graphs and equations to model real situations.

Success criteria
- ☐ I can read a value from the graph of a real situation
- ☐ I can use the rule to check the value
- ☐ I can find the rule from the graph: the starting value and the rate

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

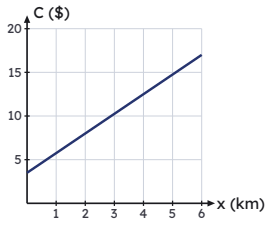
model: an equation or graph that describes a real situation
rate (gradient): the change for each 1 unit, e.g. $2.25 per km
starting value (c): the value when x = 0, e.g. a flagfall
predict: use the model to find a value

FORMULAS

y = mx + c
m = rate (change per unit)
c = starting value (x = 0)

STEPS

1. find c: the value at the start
2. find m: the change for each 1
3. write the rule
4. substitute to predict or check



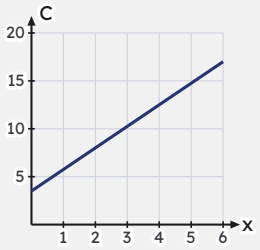
WE DO — With your teacher: copy what they write on the board.

Example 1 A rideshare costs $3.50 flagfall plus $2.25 for each kilometre.

a Table

x (km)	0	1	2	3
C ($)				

b Graph



c Write the rule for the cost C of an x km trip.

d Find the cost of a 12 km trip.

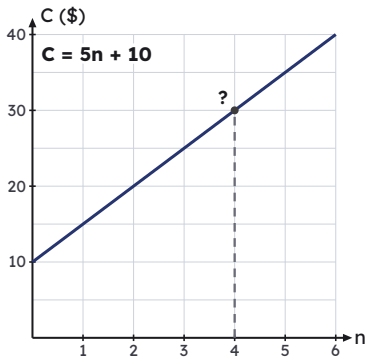
YOU DO — On your own: work down the box, one round at a time. Show your working.

2 QUESTIONS

REMEMBER Go up from the marked value to the line, then across to read it · substitute into the rule to check

ROUND 1 — Questions 1–2: read the value marked ?, then check it with the rule.

1

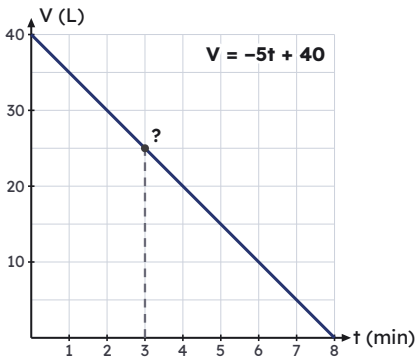


Read _____

Substitute _____

Check _____

2



Read _____

Substitute _____

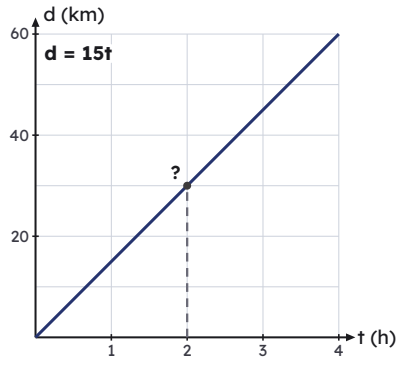
Check _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Read the scale on each axis first: each grid square may be 2, 5 or 10.

ROUND 2 Questions 3–4: read the value marked ?, then check it with the rule.

3

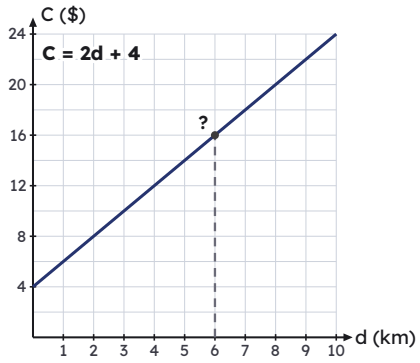


Read _____

Substitute _____

Check _____

4



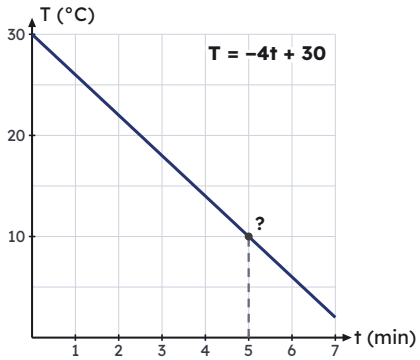
Read _____

Substitute _____

Check _____

ROUND 3 Questions 5–6: read the value marked ?, then check it with the rule.

5

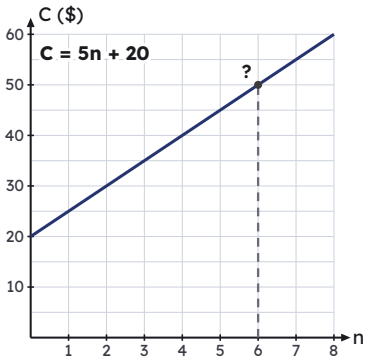


Read _____

Substitute _____

Check _____

6



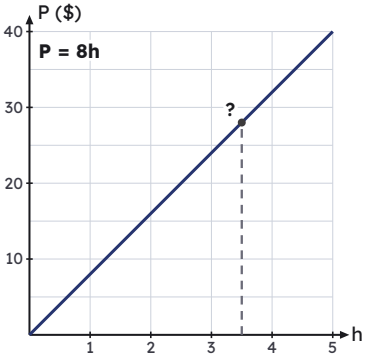
Read _____

Substitute _____

Check _____

ROUND 4 Questions 7–8: read the value marked ?, then check it with the rule.

7

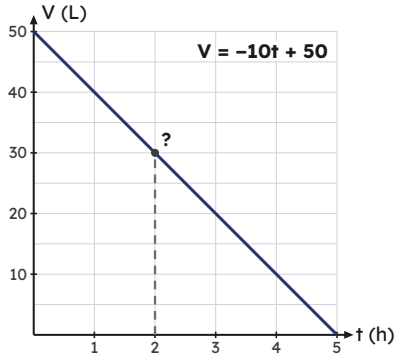


Read _____

Substitute _____

Check _____

8



Read _____

Substitute _____

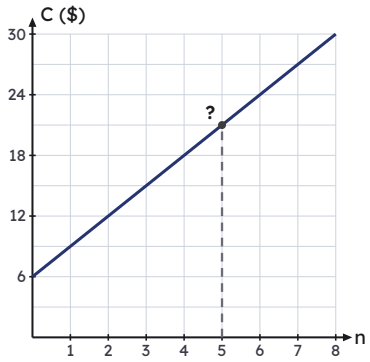
Check _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER The rule: c is where the line starts on the vertical axis; m is how much it goes up (or down) for each 1 across.

ROUND 5 Questions 9–10: find the rule from the graph, then read the value marked ?.

9

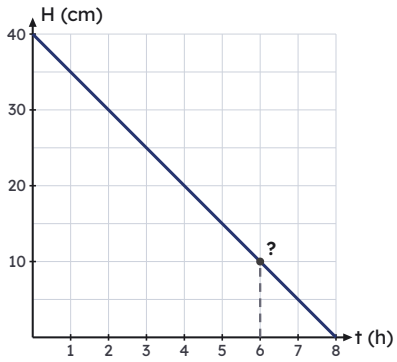


Rule _____

Read _____

Check _____

10



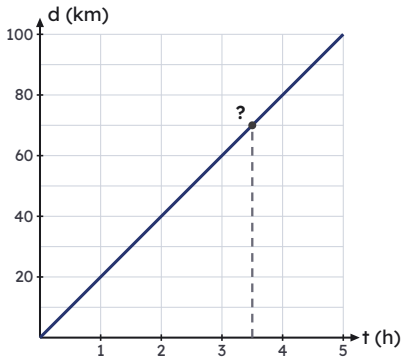
Rule _____

Read _____

Check _____

ROUND 6 Questions 11–12: find the rule from the graph, then read the value marked ?.

11

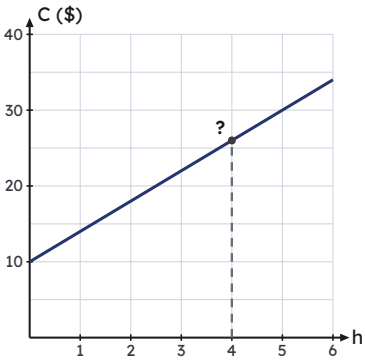


Rule _____

Read _____

Check _____

12



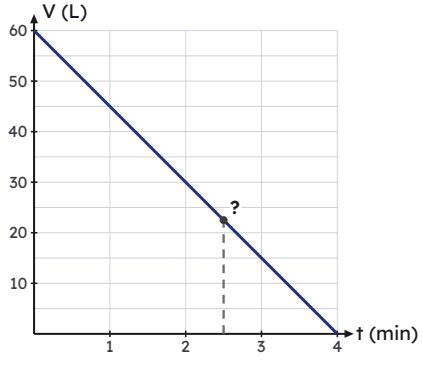
Rule _____

Read _____

Check _____

ROUND 7 Questions 13–14: find the rule from the graph, then read the value marked ?.

13

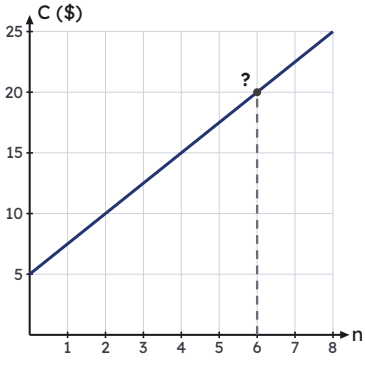


Rule _____

Read _____

Check _____

14



Rule _____

Read _____

Check _____

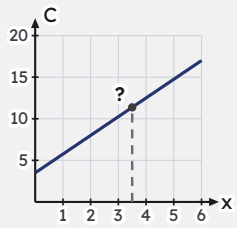
11.8 Linear modelling

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Use the rideshare model C = 2.25x + 3.5 (from Example 1).**

a Read: 3.5 km



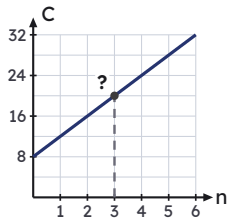
b Find the cost of a 76 km trip.

c A trip cost $12.05. How far was it?

d A trip cost $82.25. How far was it?

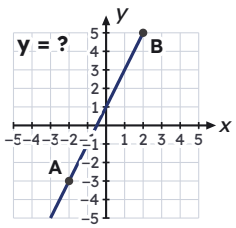
YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

1



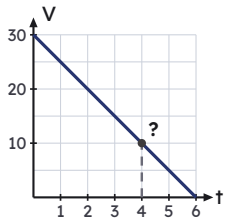
Rule _____
Read _____
Check _____

2 11.6



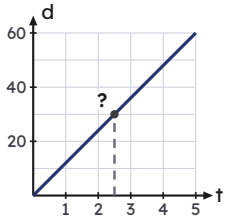
A and B _____
m and c _____
Equation _____

3



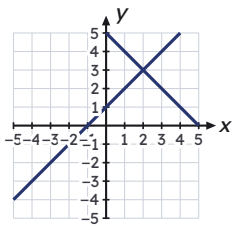
Rule _____
Read _____
Check _____

4



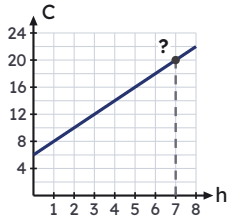
Rule _____
Read _____
Check _____

5 11.7



Read x _____
Read y _____
Point _____

6



Rule _____
Read _____
Check _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A candle is 30 cm tall and burns 2.5 cm each hour. Write a rule for its height h cm after t hours. Find its height after 5 hours, and when it burns out.

Answer _____

2 A mobile plan costs $20 a month plus 15c for each minute of calls. Write a rule for the monthly cost C of m minutes. Find the cost of 120 minutes, and how many minutes give a $50 bill.

Answer _____

3 A pool already holds 1200 L of water. Water is pumped in at 80 L a minute. Write a rule for the volume V after t minutes, and find how long until it holds 5000 L.

Answer _____

11.9 Graphing parabolas

EASY

Learning intention We are learning to graph parabolas of the form $y = ax^2 + c$ and describe their features.

Success criteria
- ☐ I can complete a table of values for $y = ax^2 + c$
- ☐ I can plot the points and join them with a smooth curve
- ☐ I can give the vertex and say whether the parabola is concave up or down

NOTES — Key ideas for this lesson

TEACHER COPY

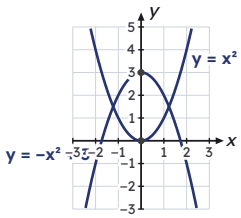
KEY TERMS

parabola: the U-shaped graph of $y = ax^2 + c$
vertex: the turning point: (0, c) here
concave up: opens upward (∪): $a > 0$
concave down: opens downward (∩): $a < 0$
axis of symmetry: the line $x = 0$: each side is a mirror image
wider / narrower: a closer to 0 is wider

FORMULAS

$y = ax^2 + c$
vertex (0, c)
$a > 0$: concave up; $a < 0$: concave down
bigger a (ignoring the sign): narrower

STEPS 1 square x first: $(-2)^2 = 4$
2 then × a, then + c 3 plot the points
4 join them with a smooth curve



WE DO — With your teacher: copy what they write on the board.

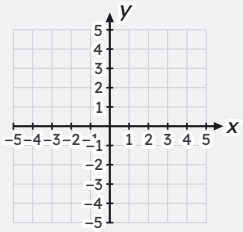
Example 1 For $y = x^2 + 1$:

a Table

$y = x^2 + 1$

x	−2	−1	0	1	2
y					

b Graph



c Give the vertex. Is it concave up or down?

d Which is wider: $y = x^2 + 1$ or $y = 3x^2 + 1$?

YOU DO — On your own: work down the box, one round at a time. Show your working.

2 QUESTIONS

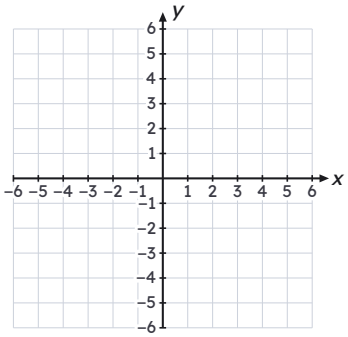
REMEMBER Square x first: $(-2)^2 = 4$ · then multiply by a · then add c · join the points with a smooth curve, not a ruler

ROUND 1 Questions 1–2: complete the table, graph the parabola, then describe it.

1

$y = x^2$

x	−2	−1	0	1	2
y					



y-values _____

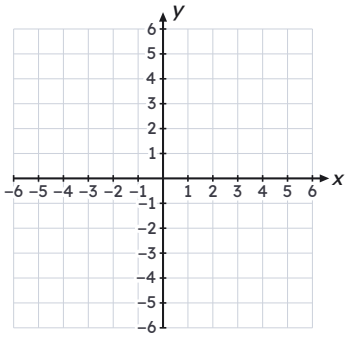
Vertex _____

Up or down _____

2

$y = x^2 + 2$

x	−2	−1	0	1	2
y					



y-values _____

Vertex _____

Up or down _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

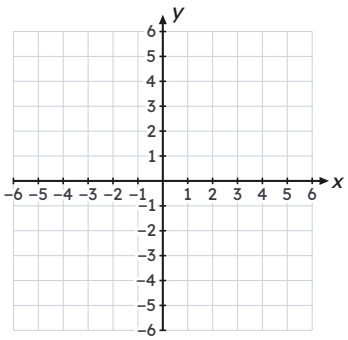
REMEMBER a negative: the parabola is upside down (∩) · the vertex is (0, c)

ROUND 2 Questions 3–4: complete the table, graph the parabola, then describe it.

3

$y = x^2 - 2$

x	−2	−1	0	1	2
y					



y-values _____

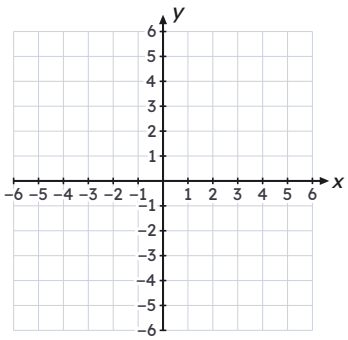
Vertex _____

Up or down _____

4

$y = -x^2$

x	−2	−1	0	1	2
y					



y-values _____

Vertex _____

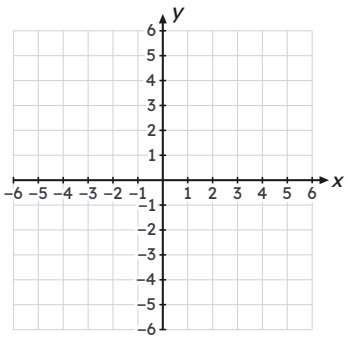
Up or down _____

ROUND 3 Questions 5–6: complete the table, graph the parabola, then describe it.

5

$y = -x^2 + 3$

x	−2	−1	0	1	2
y					



y-values _____

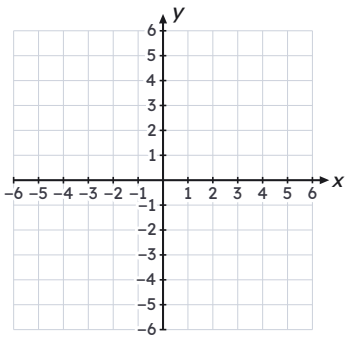
Vertex _____

Up or down _____

6

$y = x^2 - 4$

x	−2	−1	0	1	2
y					



y-values _____

Vertex _____

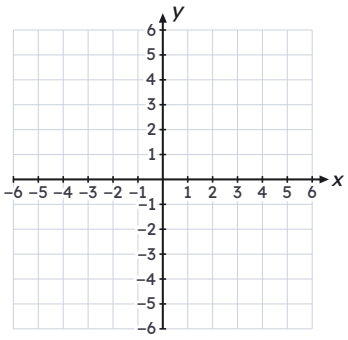
Up or down _____

ROUND 4 Questions 7–8: complete the table, graph the parabola, then describe it.

7

$y = -x^2 + 2$

x	−2	−1	0	1	2
y					



y-values _____

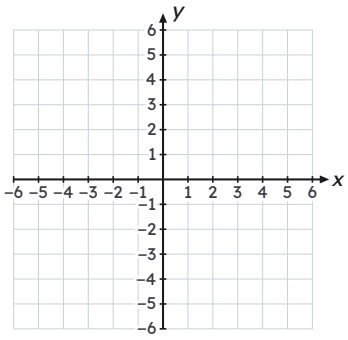
Vertex _____

Up or down _____

8

$y = x^2 - 1$

x	−2	−1	0	1	2
y					



y-values _____

Vertex _____

Up or down _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

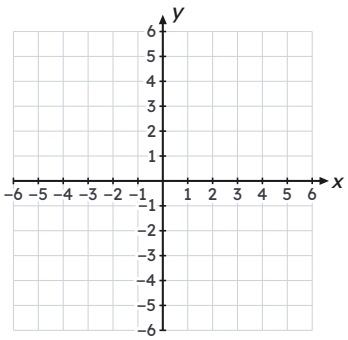
REMEMBER $y = 2x^2$: square first, then double · $y = \frac{1}{2}x^2$: square first, then halve

ROUND 5 Questions 9–10: complete the table, graph the parabola, then describe it.

9

$y = 2x^2 - 3$

x	−2	−1	0	1	2
y					



y-values _____

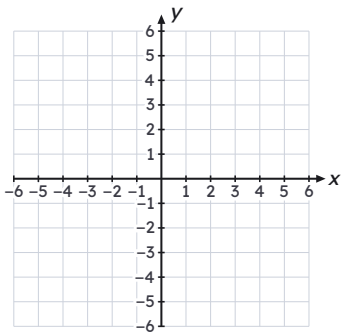
Vertex _____

Up or down _____

10

$y = -2x^2 + 4$

x	−2	−1	0	1	2
y					



y-values _____

Vertex _____

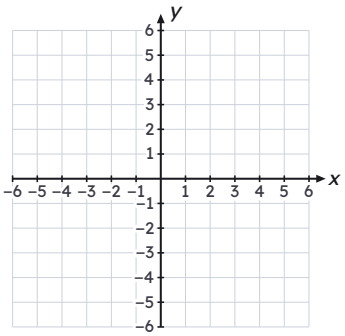
Up or down _____

ROUND 6 Questions 11–12: complete the table, graph the parabola, then describe it.

11

$y = \frac{1}{2}x^2 + 1$

x	−2	−1	0	1	2
y					



y-values _____

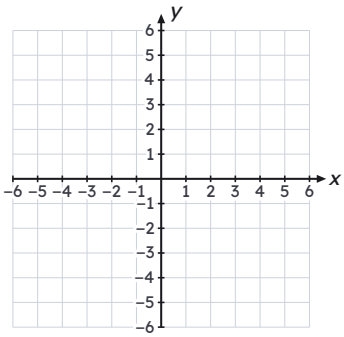
Vertex _____

Up or down _____

12

$y = -\frac{1}{2}x^2 + 2$

x	−2	−1	0	1	2
y					



y-values _____

Vertex _____

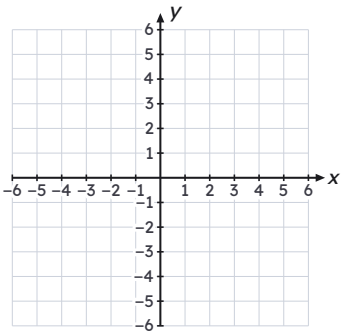
Up or down _____

ROUND 7 Questions 13–14: complete the table, graph the parabola, then describe it.

13

$y = 2x^2 - 6$

x	−2	−1	0	1	2
y					



y-values _____

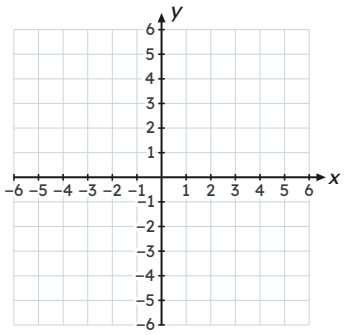
Vertex _____

Up or down _____

14

$y = -x^2 - 1$

x	−2	−1	0	1	2
y					



y-values _____

Vertex _____

Up or down _____

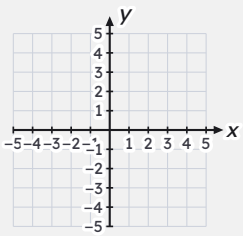
11.9 Graphing parabolas

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Graph, choose or find as shown.**

a Graph $y = -x^2 + 4$



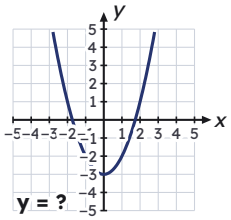
b Which is concave up with vertex (0, 0)? $y = x^2 - 1$, $y = 2x^2$, $y = -x^2$, $y = 3x^2 + 2$

c Does (3, 10) lie on $y = x^2 + 1$?

d Find the x-intercepts of $y = x^2 - 9$.

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

1 Equation

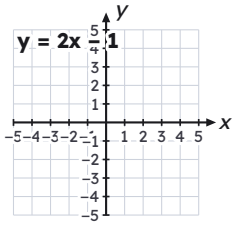


Vertex _____

Concave _____

Equation _____

2 11.5

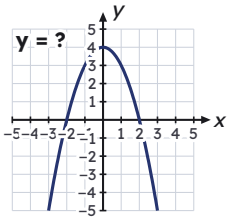


m and c _____

Two points _____

Graph _____

3 Equation

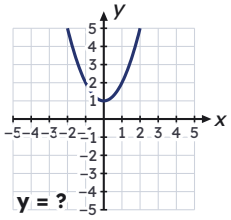


Vertex _____

Concave _____

Equation _____

4 Equation

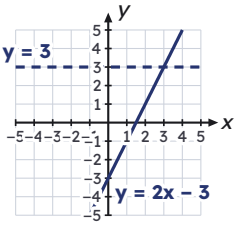


Vertex _____

Concave _____

Equation _____

5 11.7

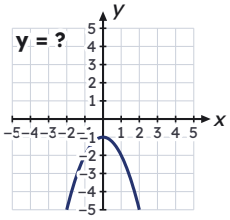


Equation _____

Read x _____

Check _____

6 Equation



Vertex _____

Concave _____

Equation _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 The arch of a bridge is modelled by $y = -0.5x^2 + 8$ (in metres). How high is the top of the arch, and how wide is the arch at ground level?

Answer _____

2 A satellite dish has the shape of $y = 0.25x^2$ (in cm). How deep is the dish 6 cm out from its centre?

Answer _____

3 Sketch $y = x^2$ and $y = x^2 + 3$ on the same axes, using $x = -2$ to 2. Describe how the second graph is related to the first.

Answer _____

11.10 Parabolas in real life

EASY

Learning intention We are learning to use parabolas to model real situations, such as falling objects.

Success criteria
- ☐ I can read a height or a time from the graph of a model
- ☐ I can substitute into the rule to check a value
- ☐ I can find the start ($t = 0$) and when the object hits the ground ($h = 0$)

NOTES — Key ideas for this lesson

TEACHER COPY

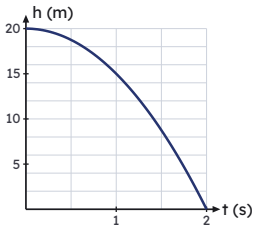
KEY TERMS

model: an equation that describes a real situation
h-intercept: the starting height ($t = 0$)
vertex: the highest or lowest point
t-intercept: when the height is 0 (on the ground)
sensible values: only times that make sense: $t \geq 0$

FORMULAS

Falling object: $h = -5t^2 + c$ (c = starting height)
Start: put $t = 0$
On the ground: put $h = 0$ and solve

STEPS 1 find the marked value on the axis
2 go to the curve, then across (or down)
3 read the value 4 check by substituting



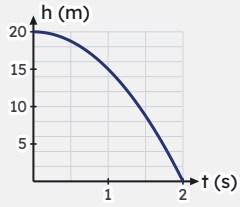
WE DO — With your teacher: copy what they write on the board.

Example 1 A stone is dropped from a cliff. Its height after t seconds is $h = -5t^2 + 20$ metres.

a Table

t	0	1	2
h			

b Graph



c Find the height after 0.5 seconds.

d When does the stone hit the ground?

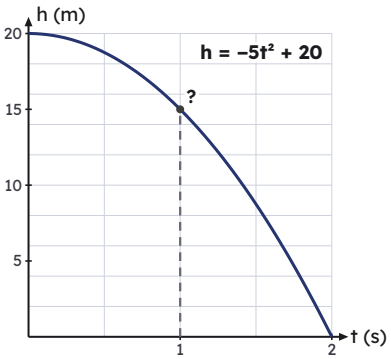
YOU DO — On your own: work down the box, one round at a time. Show your working.

2 QUESTIONS

REMEMBER Go up from the marked time to the curve, then across to read the height · square t first, then × −5, then + c

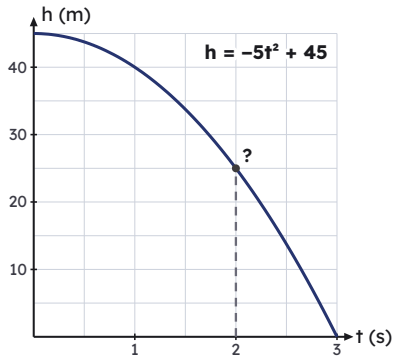
ROUND 1 Questions 1–2: read the height marked ?, then check it with the rule.

1



Read h _____
Substitute _____
Check _____

2



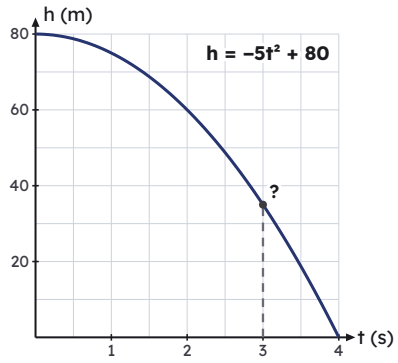
Read h _____
Substitute _____
Check _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Read the scale on each axis first: each grid square may be 2, 5, 10 or 25 metres.

ROUND 2 Questions 3–4: read the height marked ?, then check it with the rule.

3

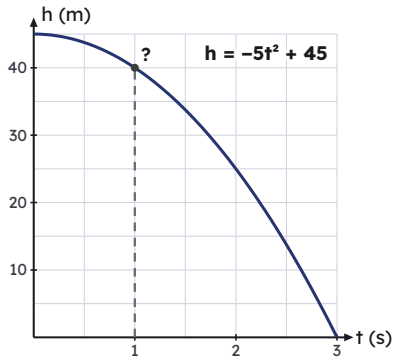


Read h _____

Substitute _____

Check _____

4



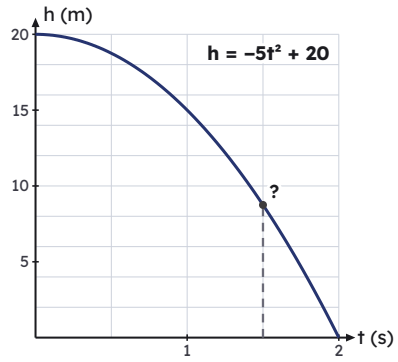
Read h _____

Substitute _____

Check _____

ROUND 3 Questions 5–6: read the height marked ?, then check it with the rule.

5

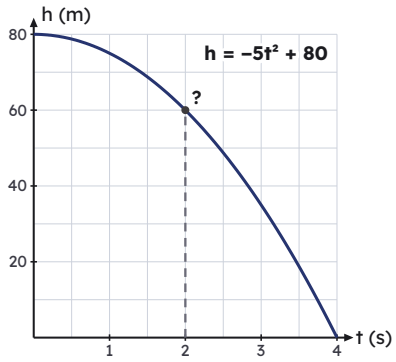


Read h _____

Substitute _____

Check _____

6



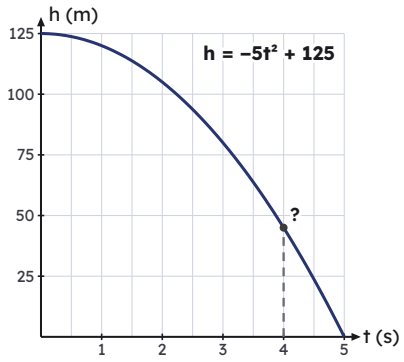
Read h _____

Substitute _____

Check _____

ROUND 4 Questions 7–8: read the height marked ?, then check it with the rule.

7

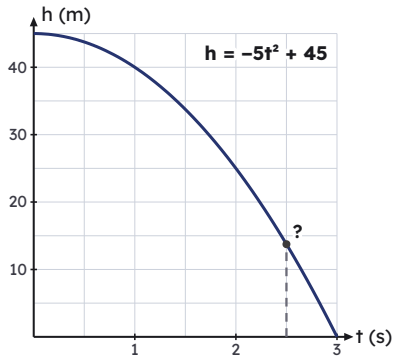


Read h _____

Substitute _____

Check _____

8



Read h _____

Substitute _____

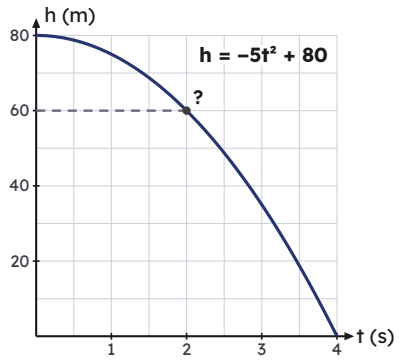
Check _____

YOU DO On your own: keep going, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Now the height is marked: go across from it to the curve, then down to read the time.

ROUND 5 Questions 9–10: read the time marked ?, then check it with the rule.

9

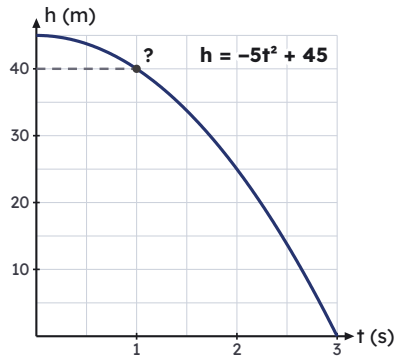


Read t _____

Substitute _____

Check _____

10



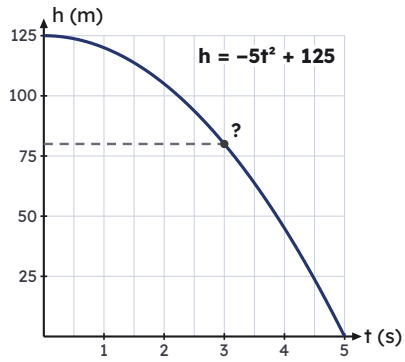
Read t _____

Substitute _____

Check _____

ROUND 6 Questions 11–12: read the time marked ?, then check it with the rule.

11

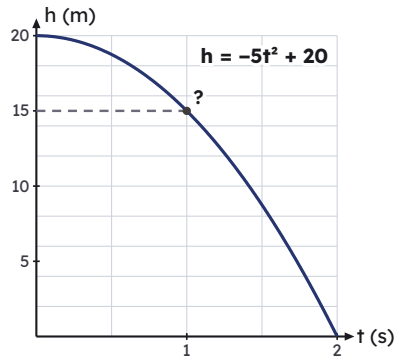


Read t _____

Substitute _____

Check _____

12



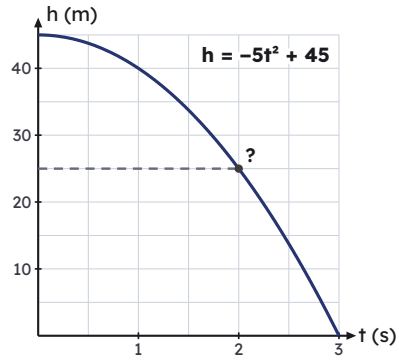
Read t _____

Substitute _____

Check _____

ROUND 7 Questions 13–14: read the time marked ?, then check it with the rule.

13

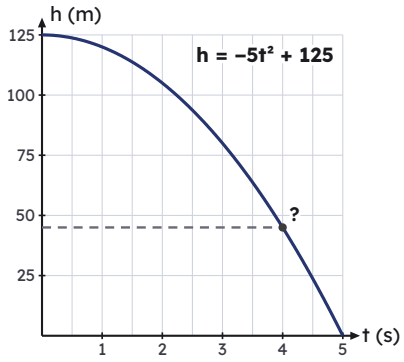


Read t _____

Substitute _____

Check _____

14



Read t _____

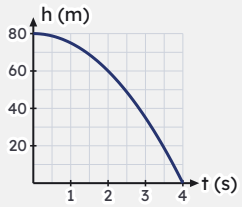
Substitute _____

Check _____

WE DO With your teacher: copy what they write on the board.

Example 2 A ball is dropped from 80 m. Its height after t seconds is $h = -5t^2 + 80$ metres.

a Graph



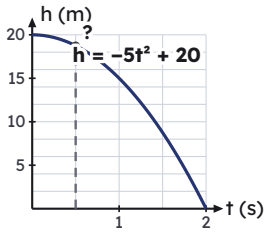
b Find the height after 2.5 seconds.

c When is the ball 35 m above the ground?

d How far does it fall in the first 2 seconds?

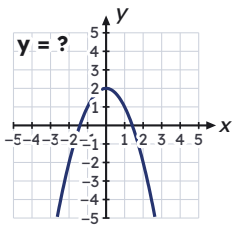
YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

1



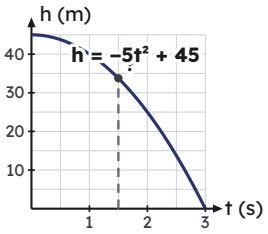
Read h _____
Substitute _____
Check _____

2 11.9



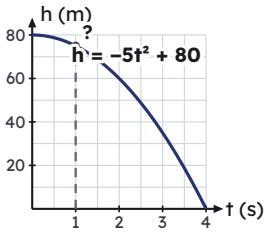
Vertex _____
Concave _____
Equation _____

3



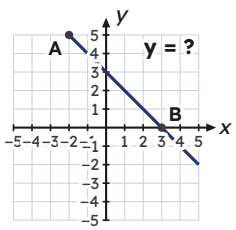
Read h _____
Substitute _____
Check _____

4



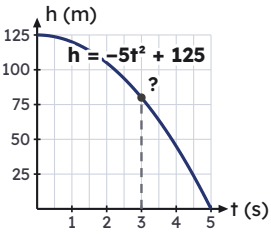
Read h _____
Substitute _____
Check _____

5 11.6



A and B _____
m and c _____
Equation _____

6



Read h _____
Substitute _____
Check _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A diver jumps from a 20 m cliff. Her height above the water after t seconds is $h = -5t^2 + 20$ metres. How long until she reaches the water? How high is she after 1.5 seconds?

Answer _____

2 A garden arch is modelled by $y = -0.25x^2 + 4$ (in metres). Find its height and its width at ground level. Can a 3 m tall cart pass under it 1.8 m from the centre?

Answer _____

3 A square photo has area $A = x^2$ cm². A frame needs a photo with an area of 50 cm². Find the side length, correct to one decimal place.

Answer _____